Inyo National Forest

Forestwide Facilities Improvement Project



Technical Specifications

November 2026

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END OF SECTION

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 00 40

PROJECT COORDINATION

PART 1 - GENERAL

1.01 SECTION INCLUDES

A. General project coordination procedures.
B. Coordination of administrative activities.
C. Conservation requirements.

1.02 RELATED REQUIREMENTS

A. Refer to Section 01 30 00 for Administrative Requirements and RFI Processing.
B. Refer to Division 01 Sections for project management, scheduling, communications, field verification, access coordination, safety, and scope clarification.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.01 PROJECT COORDINATION

A. General: Coordinate construction operations in all technical sections to ensure orderly, efficient progress.

B. Coordinate with the COR regarding site use, access, staging, traffic, and parking.

C. Coordination of Administrative Activities: Coordinate the scheduling and timing of required administrative procedures (such as submittals, progress meetings, and closeout activities) with other construction activities to avoid conflicts and ensure the orderly progress of the work.

D. Maintain communication protocols for resolving ambiguities and conflicts.

E. Coordinate administrative procedures (submittals, meetings, closeout activities) to avoid delays.

F. All work must adhere to conservation practices related to energy, water, and materials.

END OF SECTION

SECTION 01 10 00

SUMMARY OF WORK

PART 1 - GENERAL

1.01 PROJECT DESCRIPTION

A. Project Name: Forest Wide Facilities Improvement Project.

B. Project Locations include multiple district facilities and housing units across Inyo National Forest.

C. Owner: United States Department of Agriculture (USDA), Forest Service.

1.02 SUMMARY OF WORK

A. The contract includes repair, replacement, demolition, remodeling, and construction work across multiple facilities.

B. Major categories of work include:
   1. Exterior Repairs: Including roofing, siding, painting, and deck/stair repair or replacement.
   2. Interior Remodeling: Including flooring replacement, interior painting, drywall repair, and kitchen/bathroom fixture and finish upgrades.
   3. Site Improvements: Including the construction of new reinforced concrete trailer pads with associated underground utilities (water, sewer, and electrical).
   4. Specialty Work: Including the installation of pest exclusion measures, pellet stoves, and ventilation equipment.

1.03 CONTRACTOR'S USE OF PREMISES

A. Access permitted during standard hours unless otherwise approved.

B. Standard Work Hours: **7:00 AM to 5:00 PM, Monday–Friday**, excluding federal holidays.

C. Storage limited to areas designated by COR.

D. Occupancy Note (Updated): Several buildings will remain occupied. Contractor must:
   1. Always maintain safe temporary egress.
   2. Provide a 14-calendar day notice before disruptive work begins.
   3. Coordinate phasing with COR to maintain essential services.

1.04 FIELD VERIFICATION

A. Contractor must field-verify **all** dimensions prior to ordering materials.
B. No adjustments to contract value will be made for dimensional discrepancies.
C. Contractor is responsible for documenting concealed conditions and notifying COR prior to proceeding when unforeseen damage is uncovered.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION

SECTION 01 30 00

ADMINISTRATIVE REQUIREMENTS

PART 1 - GENERAL

1.01 SECTION INCLUDES

A. Pre-construction and Progress Meetings.
B. Submittal Procedures.
C. Requests for Information (RFI).
D. Closeout Submittals.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.01 PRECONSTRUCTION MEETING

A. Schedule a meeting immediately after notice to proceed:
   1. Schedule meeting at a time convenient to Government, but not later than 10 days after Notice to Proceed.
   2. Hold conference at Project site or other convenient location.

B. Attendance Required:
   1. General Contractor.
   2. Major Subcontractors.

C. Agenda:
   1. Submission of list of Subcontractors, list of Products, schedule of values, and progress schedule.
   2. Designation of personnel representing the parties to Government and Contractor including emergency off-hour contacts.
   3. Routing of correspondence.
   4. Procedures and processing of field decisions, submittals, substitutions, applications for payments, proposal requests, Change Orders, and Contract closeout procedures.
   5. Scheduling including critical work sequencing.
   6. Security and housekeeping procedures.
   7. Site access, traffic, and parking rules.
   8. Work and storage areas.
   9. Working hours.
   10. Application for payment procedures.
   11. Procedures for testing and Inspection.
   12. Procedures for maintaining record documents.

D. The Contractor shall record minutes and distribute copies within two days after the meeting to all participants, with one copy to the CO via email.

3.02 PROGRESS MEETINGS

A. The Contractor shall schedule and administer regular progress meetings at the site, at a frequency agreed upon with the COR (e.g., weekly or bi-weekly).

B. The agenda shall include a review of the project schedule, submittal status, RFIs, pending changes, and any issues affecting the work.

C. The Contractor shall record minutes and distribute them to all attendees within two days after each meeting.

3.03 SUBMITTAL PROCEDURES

A. All submittals must be contractor-generated.
B. Minimum COR review period: **10 working days**.
C. Submittals must include manufacturer requirements, compliance data, and dimensions.
D. Government will return submittals with one of the standard review stamps.

3.04 REQUEST FOR INFORMATION (RFI)

A. RFIs must include contractor's proposed solution.
B. Government response period: **10 working days.**
C. RFIs must be logged, sequentially numbered, and tracked.
D. All clarifications provided via RFI responses are **contractually binding** and incorporated into these updated specifications.

3.05 PROJECT CLOSEOUT SUBMITTALS

A. Includes as-builts, O&M manuals, warranties, and other required documentation.

END OF SECTION

SECTION 01 42 16

DEFINITIONS AND REFERENCE STANDARDS

PART 1 - GENERAL

1.01 DEFINITIONS

Wherever the following terms, or pronouns in place of them, are used in these specifications or in other contract documents, the intent and meaning are as follows:

A. **Adjustment in Contract Price:** "Equitable adjustment," as used in the Federal Acquisition Regulations.

B. **Base Course:** The layer or layers of specified or selected material of designed thickness placed on a subbase or subgrade to support a surface course.

C. **Certificate of Compliance:** A signed statement by a person with legal authority to bind a company or supplier to its product. The certificate states that the material or assemblies furnished fully comply with the requirements of the contract.

D. **Change:** "Change" means "change order" as used in the Federal Acquisition Regulations.

E. **Conduit:** A pipe for carrying fluids, wire, rope, gas, etc.

F. **Contracting Officer (CO):** The person with the authority to enter, administer, and/or terminate contracts and make related determinations and findings. The term includes certain authorized representatives of the CO acting within the limits of their authority as delegated by the CO. Authorized representatives include the Contracting Officer's Representative (COR), and Inspector.

G. **Contractor:** The individual, partnership, joint venture, or corporation undertaking the execution of the work under the terms of the contract and acting directly or through agents, employees, or subcontractors, as used in specifications and drawings.

H. **Defect:** A failure to meet a requirement with respect to a single quality characteristic.

I. **Drawings:** The documents include plan and profile sheets, plans, cross sections, diagrams, layouts, schematics, descriptive literature, illustrations, schedules, performance and test data, and similar materials showing details for construction of a facility.

J. **Equipment:** All machinery and equipment, including necessary supplies or upkeep and maintenance, as well as tools and apparatus necessary for the proper construction and acceptable completion of work.

K. **Falsework:** Any temporary construction work used to support the permanent structure until it becomes self-supporting. Falsework includes steel or timber beams, girders, columns, piles, foundations, and any proprietary equipment including modular shoring frames, post shores, and adjustable horizontal shoring.

L. **Forest Service:** The United States of America, acting through the Forest Service, U.S. Department of Agriculture.

M. **Government Land:** National Forest System lands, and other lands controlled or administered by the Forest Service or other Federal agencies.

N. **Inspector:** The Government-authorized representative designated in writing by the Contracting Officer or Contracting Officer's Representative responsible for detailed inspection.

O. **Materials:** Any substances specified for use in the construction of the project and its appurtenances.

P. **Product Data**: Printed information, such as manufacturer's installation instructions, catalog cuts, standard color charts, roughing-in diagrams and templates, standard wiring diagrams, and performance curves.

Q. **Sample**s: Partial sections of manufactured or fabricated components, cuts or containers of material, color range sets, and swatches showing color, texture, and pattern.

R. **Field Samples**: Full-sized physical examples erected on-site to illustrate finishes, coatings, or finish materials to establish standard which The Work will be judged

S. **Second Sample:** A sample taken when the initial sample indicated that the material is defective.

T. **Measurement:** Determining and expressing the quantities of work or materials.

U. **Pass:** A pass shall consist of one complete coverage of the surface.

V. **Schedule of Items:** A Schedule in the Contract that contains a listing and description of construction items, quantities, units of measure, methods of measurement, unit price, and amount.

W. **Sidewalk:** The portion of the roadway constructed primarily for pedestrian use.

X. **Specifications:** A description of the technical requirements for a material, product, or service that includes criteria for determining whether these requirements are met.

Y. **Subbase:** The prepared surface, including widening for curves, turnouts, and other areas upon which a subbase, base, or surface course is constructed.

Z. **Surface Course:** The top layer of a pavement structure, sometimes called the wearing course, usually designed to resist skidding, traffic abrasion, and disintegrating effects of climate.

1.02 REFERENCE STANDARDS

A. All work shall comply with the latest adopted edition of all applicable codes, regulations, and standards, including but not limited to:
   1. International Building Code (IBC)
   2. National Electrical Code (NEC)

3. Uniform Plumbing Code (UPC)
4. California Building Standards Code (Title 24)
5. ASTM International Standards
6. American Concrete Institute (ACI) Standards
7. Architectural Woodwork Institute (AWI) Standards

PART 2 - PRODUCTS (NOT USED)
PART 3 - EXECUTION (NOT USED)

END OF SECTION

SECTION 01 50 00

TEMPORARY FACILITIES AND CONTROLS

PART 1 - GENERAL

1.01 SECTION INCLUDES:

A. Temporary utilities (power, lighting, water).
B. Temporary sanitary facilities.
C. Temporary Controls: Barriers, enclosures, and fencing.
D. Security.
E. Vehicular access, traffic control, and parking.
F. Waste removal facilities and services.
G. Temporary fire protection.
H. Temporary heating and ventilation.

1.02 QUALITY ASSURANCE

A. Regulations: Comply with all applicable laws, codes, and regulations of authorities having jurisdiction, including but not limited to building codes, health and safety regulations, utility company regulations, and environmental protection regulations.

B. Standards: Comply with NFPA 241 "Standard for Safeguarding Construction, Alterations, and Demolition Operations" and ANSI A10 Series standards for "Safety Requirements for Construction and Demolition."

C. Electrical Service: Comply with NEMA, NECA, and UL standards. Install temporary service in compliance with NFPA 70 "National Electric Code."

1.03 TEMPORARY UTILITIES

A. The Contractor shall provide, maintain, and pay for all temporary utilities required for the work, including electricity, lighting, and water.

1.04 TEMPORARY SANITARY FACILITIES

A. Provide and maintain self-contained, single-occupant chemical toilet units for use by all construction personnel. Provide units at the time of project mobilization and service them regularly.

1.05 BARRIERS AND FENCING

A. Provide barriers and fencing to prevent unauthorized entry to construction areas, protect the public, and protect existing facilities from damage.

B. Provide appropriate warning signs (e.g., "DANGER - CONSTRUCTION AREA") to inform the public of potential hazards.

C. Heavy-duty dust containment is required in all occupied buildings during interior demolition.

D. Negative air machines with HEPA filtration are required during invasive demolition or lead-safe preparation.

1.06 **SECURITY**

A. The Contractor shall provide security and facilities to protect the work, materials, and equipment from unauthorized entry, vandalism, or theft. The Government does not provide site security.

1.07 **VEHICULAR ACCESS AND PARKING**

A. Coordinate all access routes, haul routes, and parking areas with the COR.

B. Always provide and maintain unobstructed access to fire hydrants and emergency vehicle routes.

C. The Contractor shall comply with all legal load requirements of the state and local jurisdictions. Damage to existing roadways from equipment operation shall be repaired by the Contractor at no additional cost to the Government.

D. Contractor responsible for repairing damage to roads at no additional cost.

1.08 **WASTE REMOVAL**

A. Provide waste removal facilities and services as required to maintain the site in a clean and orderly condition. Provide containers with lids and remove trash from the site periodically.

B. The Contractor is responsible for the proper removal and legal disposal of all waste generated by construction activities, including any hazardous waste.

1.09 **TEMPORARY FIRE PROTECTION**

A. Provide and maintain appropriate fire protection equipment, including portable fire extinguishers, as required by NFPA 241.

1.10 **REMOVAL OF TEMPORARY FACILITIES**

A. Remove temporary utilities and restore site prior to substantial completion.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION

SECTION 01 60 00

PRODUCT REQUIREMENTS

PART 1 - GENERAL

1.01 SECTION INCLUDES

A. General product requirements.
B. Transportation, handling, and storage.
C. Product substitution procedures.

1.02 PRODUCT DELIVERY, STORAGE, AND HANDLING

A. Deliver, store, and handle all products in accordance with the manufacturer's recommendations.

B. Deliver products to the site in an undamaged condition in the manufacturer's original sealed packaging, complete with labels and instructions.

C. Inspect all products upon delivery to ensure compliance with Contract Documents and to ensure they are undamaged.

D. Store products that are subject to damage by weather above ground and under cover in a weathertight enclosure.

PART 2 - PRODUCTS

2.01 GENERAL PRODUCT REQUIREMENTS

A. All products shall be new, unless otherwise specified.

B. Where compliance with an imposed code or standard is specified, provide a product that complies with that code or standard.

C. "American Made": Unless otherwise authorized in writing by the CO, all materials shall be from America or American Made.

PART 3 - EXECUTION

3.01 INSTALLATION OF PRODUCTS

A. Comply with the manufacturer's instructions and recommendations for the installation of all products.

B. Anchor each product securely in place, accurately located and aligned with other work.

3.02 SUBSTITUTION PROCEDURES

A. General: The products specified in this contract establish the minimum standard of quality. Substitutions will not be considered unless the specified product is unavailable.

B. Request for Substitution: The Contractor shall submit a formal Request for Substitution for any proposed change in products, materials, or equipment.
   1. Document each request with complete data substantiating the compliance of the proposed substitution with the Contract Documents, including a side-by-side comparison chart.
   2. A request for substitution constitutes a representation that the submitter has investigated the proposed product and determined that it meets or exceeds the quality level of the specified product and will provide the same warranty.

C. Review and Approval:
   1. The Contracting Officer (CO) will be the sole judge of the acceptability of any proposed substitution, and their decision is final.
   2. The COR will notify the Contractor in writing of the decision to accept or reject the request.
   3. The Contractor is responsible for any changes in the work required to incorporate an accepted substitution, at no additional cost to the Government.

END OF SECTION

SECTION 01 78 00

CLOSEOUT PROCEDURES AND SUBMITTALS

PART 1 - GENERAL

1.01 SECTION INCLUDES

A. Procedures for Substantial Completion and Final Acceptance inspections.
B. Requirements for closeout submittals, including Project Record Documents, Operation and Maintenance (O&M) Data, and Warranties.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.01 SUBSTANTIAL COMPLETION

A. Prerequisites to Inspection: Prior to requesting an inspection for Substantial Completion, the Contractor shall:
   1. Complete all work, except for minor items on a "punch list."
   2. Submit all specified warranties, operation & maintenance manuals, and final certifications.
   3. Discontinue and remove all temporary facilities from the project site.
   4. Complete final cleanup requirements.

B. Inspection Procedures:
   1. The Contractor shall submit a written request for inspection, certifying that the project is substantially complete and including a comprehensive punch list of items remaining to be completed.
   2. The COR will perform an inspection. If the work is found to be substantially complete, the COR will prepare a Certificate of Substantial Completion.

3.02 FINAL ACCEPTANCE

A. Prerequisites to Final Inspection: After Substantial Completion, the Contractor shall complete all remaining punch list items. Prior to requesting a final inspection, the Contractor shall submit:
   1. The final application for payment.
   2. A certified copy of the final inspection punch list, endorsed to show that each item has been completed.
   3. Consent of Surety to Final Payment.

B. Final Inspection: The COR will re-inspect the work to verify that all punch list items have been completed. If the work is found to be complete, the CO will issue a Certificate of Final Acceptance.

3.03 CLOSEOUT SUBMITTALS

A. Project Record Documents: Maintain a clean, undamaged set of contract documents and record all changes and "as-built" information during construction. Submit it prior to final payment.

B. Operation and Maintenance (O&M) Data: For all new equipment, compile O&M data into organized manuals. Include product data, operating instructions, maintenance procedures, and contact information for suppliers.

C. Warranties: Obtain and submit all specified warranties, executed by the manufacturers, suppliers, and subcontractors.

END OF SECTION

SECTION 01 90 00

MOBILIZATION

PART 1 - GENERAL

1.01 SCOPE

A. This work consists of preparatory work and operations necessary for the movement of personnel, equipment, supplies, and incidentals to the project site, and for all other work and operations that must be performed or that cause costs to be incurred prior to beginning work on the various items on the project site.

B. All necessary permits are to be acquired prior to commencing work.

1.02 MEASUREMENT AND PAYEMENT

A. The accepted quantities will be paid for at the contract unit price for Item 01, Mobilization.
B. Progress payments will be made as follows:
   1. The Government will pay the Contractor in the first progress payment the total premium paid by the Contractor to obtain bonds. This payment shall not be in addition to the contract price and will be included in Item 01, Mobilization.
   2. When 5 percent or more of the original contract amount is earned from other pay items, 50 percent of the amount for mobilization, or 5 percent of the original contract amount, whichever is less, will be paid.
   3. When 10 percent or more of the original contract amount is earned from other pay items, 100 percent of the amount for mobilization, or 10 percent of the original contract amount, whichever is less, will be paid.
   4. Upon completion of all work on the project, any unpaid amount for Item 01, Mobilization, will be paid.
C. The total of all payments will not exceed the original contract amount for Item 01, Mobilization.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION (NOT USED)

END OF SECTION

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 05 00

COMMON WORK RESULTS FOR EXISTING CONDITIONS

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies general administrative and procedural requirements common to all work involving existing conditions, including coordination, protection, and general site management related to existing structures and elements.

1.02 RELATED WORK

A. Refer to Division 01 for general project requirements. Refer to other Division 02 sections for specific demolition, removal, or remediation activities.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.01 PROTECTION OF EXISTING CONSTRUCTION

A. Protect all existing construction, features, and site elements designated to remain from damage, dust, debris, and contamination during the work. Implement dust control measures as necessary.

3.02 COORDINATION

A. Coordinate all work in this Division with other trades to minimize disruption to ongoing facility operations and ensure a smooth flow of work.

3.03 SITE ACCESS & SECURITY

A. Maintain secure site access and control points. Prevent unauthorized entry into work areas, especially those involving hazardous materials.

3.04 TEMPORARY UTILITIES

A. Coordinate with the COR for access to temporary power, water, and other utilities required for work in this Division.

3.05 WASTEMANAGEMENT

A. Implement a waste management plan for all non-hazardous debris generated, emphasizing recycling and proper disposal.

END OF SECTION

SECTION 02 41 00

DEMOLITION

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies the selective demolition and removal of existing interior and exterior building components, including finishes, fixtures, roofing, siding, damaged framing, and shower surrounds as required for new construction or repair work. This includes dismantling, removal, and disposal of non-hazardous elements.
B. Includes associated protection of occupants, existing utilities, landscaping, and adjacent structures.
C. Demolition work must comply with all requirements of Division 01, including dust control, HEPA filtration, and negative air where required.

1.02 RELATED WORK

A. Refer to Division 01, for Temporary Facilities and Controls (dust containment, barriers, occupancy coordination).
B. Refer to Section 02 80 00, for Discovery Protocols for Hazardous Materials.
C. Refer to Division 03, for Concrete Demolition Associated with Walkways and Trailer Pads.

1.03 FIELD CONDTIONS

A. Many buildings remain occupied, always maintain safe egress.
B. Protect existing gutters, downspouts, siding, and doors from damage. Contractor is responsible for repairing any resulting damage.
C. Protect existing utilities during excavation or demolition; use utility locating service before trenching.
D. Contractor must stop work and notify COR immediately if structural instability or hazardous materials are discovered.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.01 SELECTIVE DEMOLITION

A. Remove only those components indicated in the SOW unless otherwise directed.

B. Roofing Removal:
   1. Remove all existing roofing down to bare decking at all reroof locations.
   2. Remove underlayments, flashings, vents, drips edges, and damaged decking.
   3. Remove shingles and membranes carefully to avoid damaging gutters.

C. Bathroom Demolition:
   1. Remove water-damaged gypsum board, insulation, cove base, and deteriorated framing.

2. Remove vanities, sinks, faucets, and showers scheduled for replacement.
3. Remove old shower surrounds and pans **down to framing** (per RFI clarification).

D. Flooring Demolition:
1. Remove existing carpet, padding, vinyl, and adhesives.
2. Grind or level subfloor to correct uneven surfaces to new flooring.

E. Concrete Demolition:
1. Remove damaged sidewalks, steps, and ramps entirely where full replacement is required.
2. Remove asphalt/concrete at trailer pad areas to new grade.

3.02 HAZARDOUS MATERIALS

A. No pre-existing hazardous materials surveys exist for buildings.

B. Contractor must assume presence of Lead-based Paint (LBP) and asbestos-containing materials (ACM) for structures built before 1978.

C. Contractor must:
1. Follow EPA RRP lead-safe practices
2. Use HEPA vacuums for dust.
3. Contain work areas with plastic barriers and negative air.

D. If suspect ACM or hazardous materials are discovered:
1. Immediately stop work in that area.
2. Notify COR.
3. Await Government direction.
4. Additional abatement work is **not included in the base contract.**

3.03 UTILITY DEMOLITION

A. Disconnect utilities only where indicated.

B. Cap or make safe all abandoned utilities.

C. Coordinate shutoffs with COR.

3.04 PATCHING AND REPAIRS

A. Repair surfaces exposed by demolition to receive new work.

B. Wall and ceiling substrates must be:
1. New 5/8” moisture-resistant gypsum board in bathrooms.
2. Standard 5/8” Type X gypsum at ceilings where tiles are removed.

C. Subfloors must be:
1. Re-fastened to eliminate squeaks.
2. Leveled using approved patching compounds.
3. Re-fastened to eliminate squeaks.
4. Leveled using approved patching compounds.

D. Replace damaged studs, plates, and sheathing as necessary.

3.05 DISPOSAL AND CLEANING

A. Remove demolished materials daily.
B. Dispose of materials legally off-site.
C. Maintain clean, safe work areas for building occupants.

END OF SECTION

SECTION 02 42 00

REMOVAL AND SALVAGE OF CONSTRUCTION MATERIALS

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies procedures for the careful removal, handling, and storage of construction materials designated for salvage or recycling.

1.02 RELATED WORK

A. Refer to Section 02 41 00 for general demolition.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.01 IDENTIFICATION OF SALVAGEABLE MATERIALS

A. Prior to demolition, identify and clearly mark all materials designated for salvage by the COR.

3.02 CAREFUL REMOVAL

A. Remove salvageable materials in a manner that prevents damage and preserves their integrity for reuse or recycling.

3.03 CLEANING AND PREPARATION

A. Clean salvageable materials as directed by the COR to prepare them for storage or transport.

3.04 STORAGE

A. Store salvaged materials in a designated, secure, and weather-protected area on-site or transport them off-site to a location specified by the Owner.

3.05 RECYCLING

A. Separate and stage materials designated for recycling (e.g., metals, wood, concrete) for pickup by recycling services.

END OF SECTION

SECTION 02 80 00

DISCOVERY PROTOCOL FOR HAZARDOUS MATERIELS

PART 1 - GENERAL

1.01 SUMMARY

A. This section applies when suspected hazardous materials are encountered during demolition.

1.02 RELATED WORK

A. Refer to Sections 02 81 00, 02 82 00, 02 83 00, 02 84 00, and 02 85 00 for specific hazardous material abatement protocols.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.01 DISCOVERY PROTOCOL

A. Contractor must stop working immediately if suspected materials (asbestos, mold, requiring containment, LBP debris) are discovered.

B. Notify COR with written and photographic description.

C. Work in that zone shall not resume until Government provides written instructions.

D. Additional abatement work shall be handled via contract modification.

END OF SECTION

SECTION 02 81 00

TRANSPORTATION AND DISPOSAL OF HAZARDOUS MATERIALS

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies the requirements for the safe and lawful transportation and disposal of all hazardous materials removed from the project site, when hazardous containing materials are encountered.

1.02 RELATED WORK

A. Refer to Sections 02 80 00 through 02 85 00 for the abatement of specific hazardous materials.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.01 PACKAGING

A. Package all hazardous waste materials in approved containers, properly labeled and sealed, in accordance with DOT and EPA regulations.

3.02 MANIFESTS

A. Prepare and maintain all required manifests, shipping papers, and other documentation for hazardous waste transportation.

3.03 LICENSED TRANSPORTER

A. Utilize only licensed hazardous waste transporters for the removal of hazardous materials from the site.

3.04 APPROVED DISPOSAL FACILITIES

A. Dispose of all hazardous materials at facilities licensed and approved for the specific type of waste being disposed.

3.05 DOCUMENTATION

A. Provide the COR with copies of all manifests, waste profiles, certificates of disposal, and other relevant documentation upon completion of disposal.

END OF SECTION

SECTION 02 82 00

ASBESTOS REMEDIATION

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies the identification, safe removal, encapsulation, or enclosure of asbestos-containing materials (ACMs) from the project site, when suspected ACMs are encountered.

1.02 RELATED WORK

A. Refer to Sections 02 80 00 and 02 81 00 for general hazardous material requirements and disposal.

PART 2 - PRODUCTS

2.01 CONTAINMENT MATERIALS

A. Polyethylene sheeting, duct tape, negative air machines with HEPA filtration.

2.02 PERSONAL PROTECTIVE EQUIPMENT (PPE)

A. As required by OSHA and EPA for asbestos abatement.

2.03 ENCAPSULANTS/SEALANTS:

A. Asbestos-approved coatings for encapsulation, if specified.

PART 3 - EXECUTION

3.01 QUALIFIED PERSONNEL

A. Where applicable, all asbestos abatement shall be performed by contractor personnel who are experienced and certified in asbestos abatement safe work practices.

3.02 WORK PLAN

A. Where applicable, submit a detailed asbestos abatement work plan, including containment procedures, negative air pressure requirements, worker protection, waste handling, and air monitoring plan, for COR review and approval prior to commencing work.

3.03 CONTAINMENT

A. Where applicable, establish full containment barriers with negative air pressure where friable ACMs are being removed.

3.04 **REMOVAL PROCEDURES**

A. Where applicable, remove ACMs using wet methods and other approved techniques to minimize fiber release.

3.05 **DISPOSAL**

A. Dispose of all asbestos waste in accordance with Section 02 81 00.

END OF SECTION

SECTION 02 83 00

LEAD REMEDIATION

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies the identification, safe removal, stabilization, or encapsulation of lead-based paint (LBP) and other lead-containing materials from the project site, when suspected lead containing materials are encountered.

1.02 RELATED WORK

A. Refer to Sections 02 80 00 and 02 81 00 for general hazardous material requirements and disposal.

PART 2 - PRODUCTS

2.01 CONTAINMENT MATERIALS

A. Polyethylene sheeting, duct tape.

2.02 PERSONAL PROTECTIVE EQUIPMENT (PPE)

A. As required by OSHA and EPA for lead abatement.

2.03 ENCAPSULANTS/SEALANTS:

A. Lead-approved coatings for encapsulation or stabilization, if specified.

PART 3 - EXECUTION

3.01 QUALIFIED PERSONNEL

A. Where applicable, all lead remediation shall be performed by contractor personnel who are experienced and certified in lead-safe work practices, in accordance with EPA RRP Rule and OSHA Lead in Construction Standard.

3.02 WORKPLAN

A. Where applicable, submit a detailed lead remediation work plan, including containment procedures, worker protection, waste handling, and clearance testing plan, for COR review and approval.

3.03 CONTAINMENT

A. Establish containment barriers to prevent the spread of lead dust and debris.

3.04 REMOVAL/STABILIZATION PROCEDURES

A. Use approved methods such as wet scraping, chemical stripping, or enclosure/encapsulation to address LBP, minimizing lead dust generation.

3.05 CLEANUP AND CLEARANCE

A. Conduct thorough cleanup using HEPA vacuums and wet wiping.

3.06 DISPOSAL

A. Dispose of all lead-containing waste in accordance with Section 02 81 00.

END OF SECTION

SECTION 02 84 00

MOLD REMEDIATION

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies the assessment, containment, removal, and treatment of mold growth and mold-contaminated materials from the project site, addressing issues such as mold from previous building or roof leaks, when suspected mold containing materials are encountered.

1.02 RELATED WORK

A. Refer to Sections 02 80 00 and 02 81 00 for general hazardous material requirements and disposal.

PART 2 - PRODUCTS

2.01 CONTAINMENT MATERIALS

A. Polyethylene sheeting, duct tape, negative air machines with HEPA filtration.

2.02 PERSONAL PROTECTIVE EQUIPMENT (PPE)

A. As required for mold remediation.

2.03 BIOCIDES/FUNGICIDES

A. EPA-approved products for mold treatment, where applicable.

PART 3 - EXECUTION

3.01 QUALIFIED CONTRACTOR

A. Where applicable, all mold remediation shall be performed by contractor personnel who are experienced and certified in mold remediation protocols.

3.02 WORKPLAN

A. Where applicable, submit a detailed mold remediation work plan, including containment procedures, negative air pressure, worker protection, waste handling, and post-remediation verification plan, for COR review and approval.

3.03 CONTAINMENT

A. Where applicable, establish containment barriers to prevent the spread of mold spores to uncontaminated areas. Utilize negative air pressure and HEPA filtration.

3.04 **MOISTURE SOURCE IDENTIFICATION**

A. Identify and rectify the source of moisture causing the mold growth (e.g., repair leaks) prior to or concurrently with remediation

3.05 **REMOVAL PROCEDURES**

A. Remove and properly dispose of all mold-contaminated porous materials.

B. Clean and treat non-porous surfaces.

3.06 **POST-REMEDIATION VERIFICATION**

A. Conduct post-remediation visual inspection.

3.07 **DISPOSAL**

A. Dispose of all mold-contaminated waste in sealed bags in accordance with Section 02 81 00.

END OF SECTION

SECTION 02 85 00

HAZARDOUS MATERIAL ABATEMENT

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies the comprehensive requirements for the abatement, removal, and disposal of all hazardous materials identified on the project site. This includes, but is not limited to, asbestos, lead, and mold.

1.02 RELATED WORK

A. Refer to Section 02 81 00 for transportation and disposal specifics.

PART 2 - PRODUCTS

2.01 ABATEMENT MATERIALS

A. Containment barriers, negative air machines, HEPA filters, PPE, approved encapsulants, and cleaning agents specific to the hazardous material being abated.

PART 3 - EXECUTION

3.01 PRE-ABATEMENT SURVEY AND NOTIFICATION

A. Where applicable, the Contractor shall review the hazardous materials survey report.

B. Where applicable, notify the COR prior to commencing abatement.

3.02 QUALIFIED PERSONNEL

A. Where applicable, all hazardous material removal and abatement work shall be performed by contractor personnel that are licensed and certified for the specific type of hazardous material being abated, in accordance with all applicable federal, state, and local regulations.

3.03 SITE-SPECIFIC HEALTH AND SAFETY PLAN (HASP)

A. Where applicable**,** develop and implement a HASP for all hazardous material abatement operations, detailing procedures for worker protection, emergency response, and public notification.

3.04 CONTAINMENT AND ENGINEERING CONTROLS

A. Where applicable, establish appropriate containment barriers (e.g., critical barriers, decontamination units, negative air pressure) to prevent the release of hazardous materials to uncontaminated areas.

3.05 ABATEMENT PROCEDURES

A. Where applicable, perform abatement using approved methods to minimize disturbance and release of hazardous materials. This includes methods specific to asbestos, lead, mold, and any other identified hazardous substances.

3.06 AIR MONITORING AND CLEARANCE

A. Where applicable, conduct air monitoring as required during and after abatement.

3.07 WASTE HANDLING AND DISPOSAL

A. Where applicable, package, label, transport, and dispose of all hazardous waste materials in accordance with Section 02 81 00 and all applicable regulations.

3.08 DOCUMENTATION

A. Where applicable, maintain comprehensive records of all abatement activities, including permits, worker certifications, air monitoring reports, waste manifests, and clearance reports.

END OF SECTION

DIVISION 03 - CONCRETE

SECTION 03 10 00

CONCRETE FORMING AND ACCESSORIES

PART 1 - GENERAL

1.01 SUMMARY

A. This section includes forming and accessories for concrete trailer pads, walkways, ramps, and steps.
B. All requirements apply to full concrete replacement identified in the Schedule of Items and expanded through RFI clarifications.

1.02 RELATED WORK

A. Refer to Division 01 - General Requirements.
B. Refer to Division 31 – Earthwork

1.03 FIELD CONDITIONS

A. Contractor must field-verify all pad, walkway, and step dimensions prior to forming.
B. Protect adjacent structures, utilities, and vegetation during forming and placement.

PART 2 - PRODUCTS

2.01 MATERIALS

A. Forms shall be wood or metal, smooth, clean, free of defects.
B. Form-release agents shall not stain concrete.

PART 3 - EXECUTION

3.01 FORM INSTALLATION

A. Install forms to true lines and grades as marked in the field by COR.
B. Ensure tight joints to prevent leakage of cement paste.

END OF SECTION

SECTION 03 20 00

REINFORCING STEEL

PART 1 - GENERAL

1.01 SUMMARY

A. This section applies to the reinforcing requirements for the construction of concrete pads, ramps, stairs, and walkways.

PART 2 - PRODUCTS

2.01 REINFORCING MATERIALS

A. Reinforcing Bars:
   1. Grade 60, #4 rebar required for trailer pads.
   2. #3 or #4 permitted for walkways and steps where specified.

B. Reinforcement Accessories:
   1. Plastic-tipped chairs or spacers adequate to maintain reinforcement at mid-slab depth.

PART 3 - EXECUTION

3.01 REINFORCEMENT PLACEMENT

A. Trailer Pads:
   1. Install #4 bars at 12” on center each way, tied into a rigid grid.
   2. Match RFI-clarified structural requirement.

B. Walkways and Ramps:
   1. Provide 6x6-W1.4/W1.4 welded wire mesh or #3 rebar @16” O.C. each way.
   2. Provide continuous reinforcement at edges and transitions.

C. Maintain reinforcement cover in compliance with ACI standards.

END OF SECTION

SECTION 03 30 00

CAST-IN-PLACE CONCRETE

PART 1 - GENERAL

1.01 SUMMARY

A. This section includes cast-in-place concrete for:

1. Trailer pads (12 ft. x 25 ft.)
2. Walkway and ramp replacements (Lee Vining Ranger Station, Rock Creek House, etc.)
3. Concrete steps and landings.

1.02 QUALITY ASSURANCE

A. Concrete work shall comply with ACI 301 and 318.
B. Use a ready-mix supplier experienced with high-altitude freeze-thaw environments.

1.03 FIELD CONDITIONS

A. Work must not proceed during freezing conditions unless protective measures are used.
B. Subgrade must be compacted and stable prior to concrete placement.

PART 2 - PRODUCTS

2.01 CONCRETE MATERIALS

A. Compressive Strength: Minimum 4,000 psi at 28 days.
B. Air Entraining: Required for all exterior concrete to reduce freeze-thaw damage.
C. Water, aggregates, cement, and admixtures shall comply with ASTM standards.

2.02 SLAB THICKNESS & STRUCTURAL REQUIREMENTS

A. Trailer Pads:
   1. Minimum 6-inch uniform thickness required.
   2. Place a minimum 4-ich compacted ABC sub-base.

B. Walkways, Steps, and Ramps:
   1. Minimum 4-ich concrete thickness unless otherwise shown.
   2. Ramps must comply with ADA slope ratios and landings.

2.03 FINISHES

A. All exterior concrete surfaces shall receive a **broom finish**, perpendicular to direction of travel.

B. Stamped concrete (Rock Creek Barracks East):
   1. Required when existing stamped concrete is being replaced.
   2. Must match existing pattern and integral color.

3. Apply sealer as recommended by manufact

PART 3 - EXECUTION

3.01 SUBGRADE PREPARATION

A. Subgrade must be:
   1. Compacted to **95% maximum dry density** (trailer pads).
   2. Compacted to **90% maximum dry density** (walkways/ramps)
   3. Free of organic material.

B. Contractor must remove unsuitable soils and haul offsite if directed.

3.02 CONCRETE PLACEMENT

A. Coordinate with COR for **inspection of forms, reinforcement, and subgrade** prior to concrete placement.
B. Consolidate concrete to eliminate voids; do not overwork surface.

3.03 CURING

A. Begin curing immediately after finishing.
B. Provide moisture curing or membrane-forming curing compound.

3.04 FIELD QUALITY CONTROL

A. Provide slump, temperature, and air content tests for each batch.
B. Perform compressive strength tests per ASTM C39.

END OF SECTION

SECTION 03 35 00

CONCRETE FINISHING

PART 1 - GENERAL

1.01 SUMMARY

A. This section applies to finishing all exterior concrete surfaces.

PART 2 - PRODUCTS

A. Non-slip broom finish is required for all pedestrian surfaces.
B. Stamped concrete coloring must be **integral color**, not surface broadcast.

PART 3 - EXECUTION

A. Apply finish consistent with adjacent existing conditions.
B. Stamped concrete must match existing pattern and layout.

END OF SECTION

DIVISION 06 – WOOD, PLASTICS, AND COMPOSITES

SECTION 06 10 00

ROUGH CARPENTRY

PART 1 - GENERAL

1.01 SUMMARY

A. Includes structural wood framing, blocking, flurring, sheathing repair, and wall reconstruction.

B. Work incorporates reinforced requirements for structural repairs identified in RFIs (e.g., Crestview Cabin 1120 leaning wall issue).

C. Applies:
   1. Wall framing repairs due to rot, deflection, or water damage.
   2. New blocking required for vanities, cabinets, wall-hung units, grab bars, shower rods.
   3. Ceiling furring for drywall conversion.

1.02 RELATED REQUIREMENTS

A. Section 07 40 00 - Siding.
B. Section 09 21 00 - Gypsum Board.
C. Section 06 40 00 - Architectural Woodwork.

1.03 FIELD VERIFICATION

A. Contractor must verify existing framing, including plumb and load bearing conditions, prior to repairs.

B. Notify COR if concealed conditions extend beyond anticipated scope.

PART 2 - PRODUCTS

2.01 LUMBER Materials

A. Structural Lumber shall be:
   1. No.2 Douglas Fir-Lach or better.
   2. Pressure-treated where in contact with concrete or soil.

B. Blocking for wall-hung vanities, cabinets, and accessories shall be:
   1. **Minimum 2x6 or larger**, continuous, securely fastened**.**

2.02 PANEL MATERIALS

A. Plywood sheathing shall match existing thickness, minimum 15/32” exterior grade.
B. For structural repairs (e.g., bowed walls), COR must specify reinforcement plates or straps.

PART 3 - EXECUTION

3.01 GENERAL FRAMING INSTALLATION

A. Install framing true, square, and in accordance with engineered repair designs when required.
B. Replace deteriorated studs, sills, top plates, and cripple studs as necessary.

3.02 SPECIAL STRUCTURAL CASE – CRESTVIEW CABIN 1120

A. Structural repairs must:
   1. Jack and re-plumb the leaning south wall as required.
   2. Install new reinforcement plates or steel straps where studs are split or bowed.
   3. Replace damaged framing components.

B. All repairs must be engineered and stamped by a California-licensed Structural or Civil Engineer.

3.03 FURRING AND SUPPORT FOR CEILING CONVERSIONS

A. Install new wood furring or framing support drywall replacing ceiling tiles.
B. Ensure level, continuous substrate for 5/8” Type X gypsum board.

3.04 BLOCKING FOR WALL-HUNG COMPONENTS

A. Install continuous blocking for:
1. Wall-hung vanities.
2. Shower rods.
3. Bathroom accessories.
4. Grab bars (if required).

END OF SECTION

SECTION 06 40 00

ARCHITECTURAL WOODWORK

PART 1 - GENERAL

1.01 SUMMARY

A. This section includes fabrication and installation of cabinets, vanities, countertops, and associated woodwork.

1.02 QUALITY ASSURANCE

A. All fabrications must comply with the Architectural Woodwork Institute (AWI), **Architectural Woodwork Standards (AWS).**
B. Cheap or non-rated cabinetry is **not permitted**.

1.03 SUBMITTALS

A. Shop drawings indicating dimensions, materials, and configurations.
B. Sample of laminates, finishes, and countertop materials.

PART 2 - PRODUCTS

2.01 CABINET CONSTRUCTION

A. Cabinet Boxes must be:
   1. **¾" hardwood veneer plywood**, minimum.
   2. Industrial-grade melamine permitted only for drawer boxes or shelves.

B. Face frames, doors, and drawers must be solid wood (Poplar or Maple), factory primed.

2.02 VANITY TOPS

A. Provide solid surface or cultured marble with integrated bowl.
B. Include a 4" backsplash.
C. Approved colors selected by Government during submittals.

2.03 COUNTERTOPS

A. Countertops must be:
   1. High-Pressure Decorative Laminate (HPL).
   2. Substrate: Minimum **1-1/8" moisture-resistant particleboard or plywood**.

B. Post-formed seamless edges unless island configuration requires square edge.

PART 3 - EXECUTION

3.01 INSTALLATION OF CASEWORK

A. Install level, plumb, and secure.
B. Protect existing cabinets during countertop replacement.

3.02 VANITY REPLACEMENT

A. Replace vanities **in kind** (wall-hung replaced with wall-hung; floor-standing replaced with floor-standing).
B. Provide blocking where required.
C. Patch wall and floor surfaces exposed during removal.

3.03 COUNTERTOP INSTALLATION

A. Provide accurate cutouts for sinks and plumbing.
B. Seal all exposed substrate edges.
C. Install sinks and faucets as part of coordinated plumbing work.

END OF SECTION

SECTION 06 70 00

PLASTIC FABRICATIONS

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for composite decking materials and similar manufactured plastic items.

PART 2 - PRODUCTS

A. Composite Decking:
   1. Required where deck replacement occurs (e.g., Crestview Cabin 1122, 1012 Lee Vining House).
   2. Boards must include 20-mil wear layer, capped composite profile.

B. Fasteners:
   1. Hidden fasteners or color-matched screws.

PART 3 - EXECUTION

A. Install per manufacturer requirements.
B. Ensure adequate ventilation below decking.

END OF SECTION

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 27 00

AIR BARRIERS

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for weatherstripping at doors and windows to reduce air leakage and improve energy efficiency.

1.02 SUBMITTALS

A. Product Data: For each type of weatherstripping and sealant material.

PART 2 - PRODUCTS

2.01 WEATHERSTRIPPING

A. Compression Seals (for Door Frames):
   1. Shall be a durable, kerf-in or surface-mounted compression-type seal made of EPDM rubber or thermoplastic elastomer (TPE), designed to provide a continuous seal around the door perimeter.

B. Adhesive-Backed Seals (for Windows/Secondary Applications):
   1. Shall be a high-density, closed-cell foam or EPDM rubber tape with a high-performance, long-lasting adhesive backing. For use on window sashes or other locations where compression seals are not feasible.

C. Door Bottoms and Sweeps:
   1. Shall be an adjustable surface-mounted or U-shaped slide-on-door bottom with a flexible vinyl or brush seal to provide a tight seal at the threshold.

2.02 PERIMETER SEALANTS

A. Exterior Perimeter Sealant:
   1. Shall be a high-performance, exterior-grade sealant conforming to ASTM C920, Class 25, Grade NS. Material shall be paintable and suitable for providing a continuous, weatherproof seal between window/door frames and the adjacent wall construction (siding, trim).

2.03 FOAM SEALANTS (FOR ROUGH OOPENINGS)

A. Window and Door Foam Sealant:
   1. Shall be a low-pressure, low-expansion, AAMA-approved polyurethane foam sealant specifically formulated for insulating and sealing the gap between window/door frames and the rough opening. Foam shall not bow or deform frames.

PART 3 - EXECUTION

3.01 PREPARATION

A. Examination: Before installation, inspect all door and window frames to ensure they are sound and in a suitable condition to receive new seals.

B. Removal: Completely remove all old, deteriorated, or failed weatherstripping and sealant materials.

C. Surface Cleaning: Thoroughly clean all surfaces that will receive new seals or sealants. Surfaces must be free of dust, dirt, grease, oil, and any loose material. Use solvents for cleaning only as recommended by the sealant or adhesive manufacturer. Ensure surfaces are completely dry before installation.

3.02 INSTALLATION

A. General: Install all air barrier components in strict accordance with the manufacturer's written instructions to form a continuous and complete air seal.

B. Weatherstripping:
   1. Install weatherstripping around the full perimeter of all exterior doors and operable windows as specified in the SOW.
   2. Ensure seals are continuous at corners and provide a complete seal when the door or window is closed.
   3. The final installation shall not interfere with the normal, smooth operation of the door or window.

C. Exterior Perimeter Sealant:
   1. For joints wider than 1/4 inch, install compatible backer rod to control sealant depth.
   2. Apply sealant in a continuous, uniform bead without gaps or voids.
   3. Tool the sealant immediately after application to ensure full contact with joint surfaces and to create a smooth, concave finish.

D. Window and Door Foam Sealant:
   1. Apply low-expansion foam sealant into the gap between the window/door frame and the rough opening.
   2. Apply in accordance with the manufacturer's instructions. Do not overfill the cavity, to prevent bowing or deformation of the window or door frames.

3.03 FIELD QUALITY CONTROL

A. Visually inspect all installed seals and sealants to ensure they are continuous, properly tooled, and free of gaps.

B. After installation, operate a representative number of doors and windows to verify that they open, close, and lock smoothly without interference from the new weatherstripping.

3.04 CLEANING

A. Immediately remove any excess sealant or adhesive from adjacent surfaces.

B. Upon completion, clean the work area and remove all scrap materials and debris.

END OF SECTION

SECTION 07 30 00

ROOFING

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for the repair or full replacement of existing steep-slope roofing systems across multiple buildings.

B. Major updates from RFIs include:
   1. Mandatory architectural laminated shingles (no 3-tab allowed).
   2. Mandatory Ice & Water Shield applications.
   3. Underlayment requirements for low-slope roofs (2:12–4:12).
   4. Required wind ratings and algae protection.
   5. Dark brown drip edge requirement for all reroofed buildings.

1.02 RELATED REQUIREMENTS

A. Section 06 10 00 - Rough Carpentry
B. Section 07 62 00 - Sheet Metal Flashing and Trim.
C. Section 07 54 00 - Low-Slope Roofing Systems.
D. Section 07 70 00 - Roof Specialties.

1.03 SUBMITTALS

A. Product Data: For each component of the roofing system, including roofing material, underlayment, ice and water shield, and flashing.

B. Samples: Submit samples of the specified roofing material, showing the full range of available colors for selection by the COR.

C. Manufacturer's Instructions: Submit complete installation instructions.

D. Warranty: Submit sample copies of the manufacturer's material warranty (lifetime limited, wind, algae resistance etc.) and the installer's workmanship warranty.

1.04 WARRANTY REQUIREMENTS

A. Provide **Lifetime Limited Warranty** for laminated architectural shingles.
B. Provide **minimum 20-year non-prorated protection period** covering material and labor.
C. Wind warranty: **110 mph minimum**, upgradeable to **130 mph** with enhanced fastening pattern.
D. Algae resistance warranty: **Minimum 10–15 years.**

1.05 QUALITY ASSURANCE

A. Installer Qualifications: Installer shall have a minimum of five years of experience in installing roofing systems of the type specified.

PART 2 - PRODUCTS

2.01 PRIMARY ROOFING MATERIAL

A. The primary roofing material shall be as specified in the detailed Statement of Work (SOW). Options include, but are not limited to:
   1. Architectural Asphalt Shingles
      a) Shall be premium laminated architectural fiberglass asphalt shingles.
      b) Shall meet ASTM D3462
      c) Shall meet ASTM D3161 Class F or Class H wind resistance.
      d) Include copper-infused algae-resistant granules.

   2. Standing Seam Metal Roofing: Minimum 24-gauge steel or equivalent aluminum with a factory-applied, high-performance coating (e.g., Kynar 500 / Hylar 5000).

2.02 UNDERLAYMENT AND MOISTURE BARRIERS

A. Ice and Water Shield:
   1. Self-adhering, rubberized asphalt membrane, minimum 40 mils thick. Shall be used at all eaves, valleys, and around all roof penetrations (vents, chimneys, skylights).
   2. For standard roof slopes (<4:12), apply self-adhering ice & water shield from eaves to 36 inches inside heated wall line.
   3. Full coverage ice & water shield is mandatory for low-slope roofs (2:12 to 4:12).

B. Underlayment:
   1. Synthetic Underlayment: High-strength, non-woven polypropylene or similar synthetic material, as recommended by the primary roofing manufacturer.
   2. Asphalt-Saturated Felt: ASTM D226, Type II (30), non-perforated.
   3. For standard roof slopes (>4:12), synthetic underlayment shall be applied over remaining deck.
   4. Felt or synthetic-only underlayments are prohibited for low-slope roofs (2:12 to 4:12).
   5. Full deck coverage for 2:12 - 4:12 slopes.

2.03 FLASHING AND TRIM

A. Metal Flashing: Minimum 24-gauge galvanized steel or 0.019-inch aluminum, with a finish compatible with the primary roofing material. All flashing shall be new, dark brown or as directed by COR, and prefinished.

B. Components: Shall include, but not be limited to:
   1. Drip Edge: To be installed at all eaves and rakes.
   2. Step Flashing: For all wall-to-roof intersections.
   3. Valley Flashing: Open or closed valley flashing as appropriate for the roof type.
   4. Vent Pipe Flashing: New boots for all plumbing vents.

2.04 FASTENERS

A. All fasteners shall be as recommended by the roofing manufacturer for the specific system being installed.

B. For Asphalt Shingles: Use hot-dip galvanized steel or stainless-steel roofing nails of sufficient length to penetrate the roof deck by at least 3/4 inch.

C. For Metal Roofing: Use concealed clips and gasketed screws designed for the specific metal roofing system to prevent leaks and allow for thermal movement.

D. For high wind zones, install six (6) fasteners per shingle.

2.05 **ROOF VENTILATION**

A. Ridge Vents: Continuous ridge vent system compatible with the primary roofing material.

B. Soffit Vents: Continuous or individual soffit vents to provide intake ventilation.

C. System Requirement: The ventilation system (ridge and soffit vents) shall be balanced and meet or exceed local building code requirements for net free vent area.

2.06 **ROOF DECKING REPLACEMENT**

A. Contractor must include cost for up to three (3) sheets of plywood replacement per building within base bid.
B. Additional deck replacement handled via modification after COR verification.

PART 3 - EXECUTION

3.01 **EXAMINATION AND PREPARATION**

A. Site Protection: Before beginning work, protect the building's siding, windows, landscaping, and surrounding areas from potential damage from falling debris. Use tarps or other protective coverings.

B. Deck Inspection: After the complete removal of the existing roofing system, thoroughly inspect the roof deck for rot, delamination, or damage. Report any unsatisfactory conditions or areas requiring repair or replacement to the COR in writing. Do not proceed with new roofing installation until the deck is sound.

3.02 **DEMOLITION (FOR FULL REPLACEMENT)**

A. Completely remove all existing roofing materials, underlayment, flashing, and fasteners down to the bare roof deck. Ensure all old nails are removed or hammered flat.

B. Sweep the roof deck clean of all dirt and debris before installing any new materials.

3.03 **INSTALLATION**

A. General: Install all roofing system components in strict accordance with the manufacturer's written instructions and all applicable building codes. The manufacturer's instructions shall be the minimum standard; where codes are more stringent, they shall govern.

B. System Installation Sequence: Install roofing system components in the following sequence to ensure a weather-tight installation:

1. Drip Edge: Install drip edge flashing at all eaves (under the ice and water shield) and rakes (over the underlayment).
2. Ice and Water Shield: Install self-adhering ice and water shield at all eaves, extending at least 24 inches inside the interior plane of the exterior wall. Install at all valleys and around all roof penetrations (vents, chimneys, etc.).
3. Underlayment: Install underlayment over the entire remaining roof deck, overlapping courses in a weatherboard fashion as per manufacturer's instructions.
4. Flashing: Install all new metal flashing, including step flashing, valley flashing, and vent pipe boots. Ensure all flashing is properly integrated with the underlayment and roofing material to direct water away from openings.
5. Primary Roofing Material: Install the primary roofing material (shingles or metal panels) starting from the eaves and working towards the ridge, following the manufacturer's specific requirements for alignment, fastening, and exposure.
6. Ventilation and Caps: Install ridge caps and ventilation systems as the final step, ensuring a complete and weather-tight finish.

3.04 REPAIR

A. For roof repairs as specified in the SOW, carefully remove only the damaged or leaking sections of the roof.

B. Integrate new materials (decking, underlayment, flashing, roofing) with the existing system to match as closely as possible and create a permanent, watertight seal.

3.05 PROTECTION OF EXISTING GUTTERS

A. Most gutters remain in place; contractor must protect them.
B. Replace only the downspouts indicated in SOW**.**

3.06 FIELD QUALITY CONTROL

A. The COR reserves the right to inspect the work at any stage of the installation.

B. Inspect fastener placement to ensure compliance with the manufacturer's nailing patterns.

C. Verify that flashing is installed correctly at all transitions and penetrations.

3.07 CLEANUP

A. Keep the project site clean daily, removing all roofing debris from the ground and gutters.

B. Upon completion, perform a thorough final cleanup of the entire site. Use a magnetic sweeper to remove fallen nails from lawns, driveways, and walkways.

C. Leave the site in a clean, undamaged condition.

3.08 PROTECTION

A. Protect the newly installed roof from damage by subsequent construction activities. The Contractor is responsible for repairing any damage to the new roof that occurs before final project acceptance.

END OF SECTION

SECTION 07 40 00

SIDING

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for the repair or full replacement of exterior siding systems, including the weather-resistive barrier, trim, and accessories.

B. Only minor patching included in painting line items; full board replacement covered in this section.

1.02 RELATED SECTIONS

A. Section 09 90 00 - Painting and Coating
B. Section 06 10 00 - Rough Carpentry.

1.03 SUBMITTALS

A. Product Data: For siding material, weather-resistive barrier, trim, and fasteners.

B. Samples: Submit samples of the specified siding material, showing the full range of available colors and textures for selection by the COR.

1.04 QUALITY ASSURANCE

A. Installer Qualifications: Installer shall have a minimum of five years of experience in installing siding systems of the type specified.

PART 2 - PRODUCTS

2.01 PRIMARY SIDING MATERIAL

A. The primary siding material shall match the profile, dimensions, exposure, and thickness of existing siding.

B. Siding material Options include, but are not limited to:
   1. Stucco System (Three-Coat): A complete system consisting of metal lath, a scratch coat, a brown coat, and a finish coat, all conforming to ASTM C926. The finish coat shall have an integral color and texture as selected by the COR.
   2. Fiber Cement Siding: Conforming to ASTM C1186. Type (e.g., lap siding, panels) and texture (e.g., smooth, wood grain).
   3. Wood Siding: Solid wood naturally resistant to decay, such as Cedar or Redwood. Grade and profile (e.g., lap, tongue-and-groove).

2.02 WEATHER-RESISTANT BARRIER (WRB)

A. General: A WRB shall be installed over all wall sheathing prior to the installation of the siding system.

B. Building Paper: Minimum two layers of Grade D building paper or one layer of #15 asphalt-saturated felt (ASTM D226).

C. Synthetic House Wrap: A non-woven, breathable synthetic house wrap, as approved by the siding manufacturer.

D. Flashing Tape: Self-adhering, flexible flashing tape for sealing around all window and door openings.

2.03 TRIM AND ACCESSORIES

A. Trim Material: Shall be compatible with the selected siding material.
   1. For Fiber Cement or Wood Siding: Use composite (PVC), fiber cement, or wood trim for all corners, window/door surrounds, and fascia.
   2. For Stucco Systems: Use galvanized steel or PVC accessories, including corner beads, casing beads (to terminate stucco at openings), and weep screeds (at the base of all walls).

B. Vented Components: Where specified, provide vented soffit panels or other ventilation accessories compatible with the siding system.

2.04 FASTENERS

A. All fasteners shall be corrosion-resistant (hot-dip galvanized or stainless steel) and shall be of the type and size recommended by the siding manufacturer for the specific system and substrate.

B. For Fiber Cement/Wood Siding: Use siding nails or screws of sufficient length to penetrate the wall studs by at least 1-1/4 inches.

C. For Stucco Lath: Use galvanized staples or screws as required by code to securely fasten the lath to the wall framing.

PART 3 - EXECUTION

3.01 EXAMINATION AND PREPARATION

A. Site Protection: Protect adjacent surfaces, windows, and landscaping from damage during siding installation.

B. Substrate Inspection: After the complete removal of existing siding, thoroughly inspect the wall sheathing for any rot, delamination, or damage. Verify that the wall surface is flat and suitable for siding installation. Report any unsatisfactory conditions to the COR in writing. Do not proceed until the substrate is sound.

3.02 INSTALLATION

A. General: Install all siding system components in strict accordance with the manufacturer's written instructions and all applicable building codes. The manufacturer's instructions shall be the minimum standard; where codes are more stringent, they shall govern.

B. Weather-Resistive Barrier (WRB) Installation:
   1. Install the WRB over the entire wall sheathing surface, starting from the bottom and overlapping upper courses over lower courses in a weatherboard fashion by a minimum of 6 inches.
   2. Overlap vertical seams by a minimum of 6 inches. Tape all seams with manufacturer-approved tape.
   3. Integrate the WRB with window and door flashing tapes to create a continuous, shingled drainage plane that directs all moisture out and away from the wall assembly.

C. Trim and Accessory Installation:
   1. Install all corner boards, window/door trim, and other trim accessories before installing the main siding field.
   2. For stucco, install weep screed at the base of all walls (held a minimum of 4 inches above earth) and casing beads at all terminations (windows, doors, soffits).

D. Primary Siding Installation:
   1. For Lap Siding (Fiber Cement/Wood):
      a. Maintain a minimum 2-inch clearance between siding and roofing, decks, or walkways, and a minimum 6-inch clearance to grade.
      b. Install siding with the exposure and overlap specified by the manufacturer.
      c. Securely fasten siding to wall studs. Do not over-drive fasteners.
   2. For Stucco Systems:
      a. Install metal lath over the WRB in accordance with ASTM C1063.
      b. Apply the three-coat system (scratch, brown, finish) in accordance with ASTM C926.
      c. Allow for proper curing time between each coat as required by the manufacturer and industry standards.
      d. Apply the finish coat to achieve a uniform color and texture, free of cold joints or blemishes.

3.03 REPAIR

A. For siding repairs as specified in the SOW, carefully remove only the damaged sections of siding.

B. Integrate new materials with the existing system to match the original profile, exposure, and texture as closely as possible, ensuring a permanent, watertight repair.

3.04 FIELD QUALITY CONTROL

A. Verify that flashing is correctly installed at all siding penetrations, openings, and transitions.

B. Inspect fastener placement to ensure compliance with the manufacturer's nailing patterns and to prevent over-driving or under-driving.

3.05 CLEANUP

A. Keep the project site clean daily, removing all siding scraps, packaging, and debris.

B. Upon completion, perform a thorough final cleanup of the entire site, including removal of any dust or residue from adjacent surfaces.

C. Leave the site in a clean, undamaged condition.

END OF SECTION

SECTION 07 54 00

LOW-SLOPE ROOFING

PART 1 - GENERAL

1.01 SUMMARY

A. Applies to low-slope areas such as the Helitack Office Shed connector roof (1/2:12 pitch).

PART 2 - PRODUCTS

2.01 MEMBRANE SYSTEM

A. Install 2-ply or 3-ply SBS-modified bitumen, self-adhered.
B. Cap sheet must be granule-surfaced and color-matched to shingles.

2.02 WARRANTY

A. Acceptable warranty: 10-year material-only, plus contractor’s 2-year workmanship warranty.
B. Full NDL warranty is not required.

PART 3 - EXECUTION

A. Install per manufacturer’s low-slope guidelines.
B. Seal laps properly.
C. Install new perimeter metal flashings.

END OF SECTION

SECTION 07 62 00

SHEET METAL FLASHING AND TRIM

PART 1 - GENERAL

1.01 SUMMARY

A. Includes new drip edges, flashing, and trim.

PART 2 - PRODUCTS

A. Drip Edge:
   1. Material: 24-gauge galvanized steel or heavy-duty aluminum.
   2. Finish: Dark Brown prefinished enamel (mandatory).

B. Flashings: galvanized steel or aluminum, matching existing profiles.

PART 3 - EXECUTION

A. Install straight and true, with secure fasteners.
B. Ensure full integration with roofing system.

END OF SECTION

SECTION 07 70 00

ROOF SPECIALTIES

PART 1 - GENERAL

1.01 SUMMARY

A. Includes engineered snow retention systems for metal roofs.

PART 2 - PRODUCTS

A. Provide continuous snow bars or rails installed into structural framing.

PART 3 - EXECUTION

A. Install per engineered layout.
B. Match color of metal roofing.

END OF SECTION

SECTION 07 80 00

FIRE PROTECTION (HISTORIC SHAKE ROOF)

PART 1 - GENERAL

1.01 SUMMARY

A. Applies only to 1025 Lee Vining Small House.

PART 2 - PRODUCTS

A. Western Red Cedar shakes - Certi-Split Premium Grade, hand-split, resawn, edge grain.
B. Fire-retardant treated (Certi-Guard).
C. Must meet Class B with a minimum rating (Class A if combined with approved underlayment system).

PART 3 - EXECUTION

A. Match historical dimensions and spacing.
B. Coordinate with COR regarding State Historic Preservation Office (SHPO) related requirements.

END OF SECTION

DIVISION 08 - OPENINGS

SECTION 08 10 00

DOORS

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for new exterior and interior doors, frames, and all associated hardware.

B. Incorporates RFI clarifications requiring:
   1. All exterior doors to prehung units (no slab-only replacements).
   2. SFIC-compatible locksets for Forest Service Master keying.
   3. Required material types for different building locations.
   4. Interior door replacement only where indicated or where frames are out of plumb.

1.02 RELATED REQUIREMENTS

A. Division 06 - Rough Carpentry.
B. Division 09 - Painting.
C. Division 07 - Siding and Roofing.

1.03 SUBMITTALS

A. Product Data: For each type of door, frame, and piece of hardware.

B. Samples: If requested, submit samples of hardware finishes for approval.

PART 2 - PRODUCTS

2.01 EXTERIOR DOORS AND FRAMES

A. General: Exterior doors shall be provided as pre-hung units, complete with frame, hinges, threshold, grade 2 lockset with deadbolt and weatherstripping from a single manufacturer.

B. Door Material: Options include:
   1. Insulated Steel Door: Minimum 24-gauge steel skins with a polyurethane foam core, factory-primed for painting. Shall be installed in utility or service spaces.
   2. Fiberglass Door: Fiberglass composite skins with a polyurethane foam core, with a smooth or textured (wood grain) finish, suitable for painting or staining. Shall be installed in housing and barracks spaces.

C. Frames: Shall be wood or composite material, compatible with the door and suitable for exterior exposure.

2.02 INTERIOR DOORS AND FRAMES

A. General: Interior doors shall be provided as pre-hung units with frames or as door slabs for installation into existing frames, as specified in the SOW.

B. Door Material: Shall be as specified in the SOW. Options include:
   1. Hollow Core: Molded composite or hardboard, with a smooth or textured finish, factory-primed for painting. For use in standard passage applications like closets and bedrooms.
   2. Solid Core: Solid particleboard or mineral core with a wood veneer or molded composite skin. For use where sound control or a more durable door is required, such as at bathrooms or offices.

C. Frames: Shall be finger-jointed pine or other paint-grade wood.
D. Install as slabs only if existing frames are sound; otherwise use prehung units.

2.03 SPECIALITY DOORS

A. Storm Doors: Shall be an aluminum-frame storm door with interchangeable glass and screen panels, complete with all hardware including latch, closer, and wind chain.

B. Screen Doors: Shall be a wood or aluminum frame screen door with durable fiberglass or aluminum screen mesh.

2.04 HARDWARE

A. Locksets and Latchsets:
   1. Exterior Doors: Keyed entry function lockset with deadbolt, ANSI/BHMA Grade 2 minimum.
   2. Interior Doors (Bathrooms/Bedrooms): Privacy function (push-button or turn-piece lock).
   3. Interior Doors (Closets/Hallways): Passage function (non-locking).
   4. Finish: To be specified in the SOW (e.g., "Satin Nickel," "Oil-Rubbed Bronze").

B. Hinges:
   1. Material: Steel, with a finish to match the lockset.
   2. Quantity: Minimum of three (3) hinges per door.

C. Thresholds and Weatherstripping (for Exterior Doors):
   1. Threshold: Adjustable aluminum threshold with a vinyl seal.
   2. Weatherstripping: Compression-type weatherstripping on the frame and a door bottom/sweep to create a complete seal.

D. Doorstops:
   1. Provide wall-mounted or hinge-pin door stops for all interior doors to protect walls from damage.

2.05 WEATHERSTRIPPING

A. Provide new adjustable perimeter compression weatherstripping for exterior doors.
B. Provide new door sweeps.
C. Mammoth Fire Barracks requires weatherstripping at two (2) primary exterior doors (per RFI clarification).

2.06 **LOCKSETS AND KEYING**

A. All exterior locksets must be Small Format Interchangeable Core (SFIC) compatible.

PART 3 - EXECUTION

3.01 **EXAMINATION**

A. Verify Openings: Before beginning installation, verify that rough openings are plumb, level, square, and correctly dimensioned for the door units to be installed.

B. Substrate Condition: Inspect the subfloor and wall framing for any conditions that could affect the quality of the installation. Report any discrepancies or unsatisfactory conditions to the COR in writing before proceeding.

3.02 **INSTALLATION**

A. General: Install all doors, frames, and hardware in strict accordance with the manufacturer's written instructions and all applicable building codes.

B. Pre-Hung Door Units:
   1. Set pre-hung door frames plumb, level, and square within the rough opening.
   2. Shim the frame behind each hinge and at the latch side, ensuring the gaps between the door and the frame are uniform on all sides.
   3. Securely anchor the frame to the wall framing through the shims. Do not over-tighten fasteners, to avoid deforming the frame.

C. Sealing (Exterior Doors):
   1. Apply a continuous bead of sealant along the subfloor under the threshold before setting the frame.
   2. After securing the frame, insulate the gap between the frame and the rough opening with low-expansion foam sealant.
   3. Apply exterior-grade sealant around the perimeter of the exterior trim to create a complete weather seal.

D. Hardware Installation:
   1. Install all locksets, latchsets, hinges, and other hardware according to the manufacturer's templates and instructions.
   2. Ensure all hardware is securely fastened and functions correctly.

E. Specialty Door Installation:
   1. Install storm and screen doors according to the manufacturer's instructions, ensuring the door closer and latching hardware are properly adjusted for smooth operation.

F. Replace damaged trim at door openings.

3.03 **ADJUSTMENT**

A. After installation, adjust all doors to ensure they swing freely and smoothly without binding or sticking.

B. Adjust all locksets and latchsets to ensure they engage properly and securely without excessive force.

C. Verify that all exterior doors create a complete, tight seal against the weatherstripping when closed to prevent air and water infiltration. Adjust the threshold and sweep as necessary.

3.04 CLEANING AND PROTECTION

A. Keep the work area clean daily, removing all packaging, scrap materials, and debris.

B. Upon completion, clean all door and hardware surfaces of any dust, fingerprints, or construction marks.

C. Protect all installed doors and hardware from damage by subsequent construction activities until final project acceptance.

END OF SECTION

SECTION 08 11 00

METAL DOORS AND FRAMES

PART 1 - GENERAL

1.01 SUMMARY

A. Applies to insulated steel exterior doors or service-room doors.

PART 2 - PRODUCTS

A. Minimum 24-gauge insulated steel.
B. Factory primed.

PART 3 - EXECUTION

A. Install prehung units including thresholds.
B. Seal all edges and penetrations.

END OF SECTION

SECTION 08 14 00

INTERIOR WOOD DOORS

PART 1 - GENERAL

1.01 SUMMARY

A. Applies to all interior door replacements.

PART 2 - PRODUCTS

A. Bathroom doors must be solid core.
B. Bedroom and closet doors may be hollow core.
C. Match existing sizes unless otherwise indicated.

PART 3 - EXECUTION

A. Install to match existing swing and handing.
B. Replace interior casing as needed.

END OF SECTION

SECTION 08 40 00

ENTRANCE DOORS, STORM AND SCREEN DOORS

PART 1 - GENERAL

1.01 SUMMARY

A. Applies to buildings requiring storm and/or screen door replacement.

PART 2 - PRODUCTS

A. Storm Doors:
   1. Heavy-duty aluminum frame.
   2. Interchangeable glass and screen inserts.
   3. Pneumatic closer.
   4. Matching hardware.

B. Screen Doors
   1. Rigid aluminum or wood frame.
   2. Heavy-duty fiberglass or aluminum mesh.
   3. Self-closing hinges.

PART 3 - EXECUTION

A. Match historical dimensions and spacing.
B. Install plumb and secure.
C. Coordinate with primary door installation.
D. Install weatherstripping as required.

END OF SECTION

SECTION 08 50 00

WINDOWS

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for new window units, including frames, glazing, hardware, and installation materials.

B. Incorporates clarifications from RFIs regarding:
   1. Frame material (vinyl-only)
   2. Energy performance requirements
   3. Tempered glazing requirements
   4. Egress windows in bedrooms
   5. Window counts for Crestview cabins

1.02 SUBMITTALS

A. For each type of window, showing frame material, glazing specifications, NFRC energy performance ratings, and installation instructions.

B. Shop drawings verifying dimensions.

C. Color selection samples.

D. Warranty: Submit sample copies of the manufacturer's warranty.

1.03 QUALITY ASSURANCE

A. Installer Qualifications: Installer shall have a minimum of five years of experience in installing windows of the type specified.

PART 2 - PRODUCTS

2.01 WINDOWS

A. General: All windows shall be new, factory-assembled units, complete with frames, glazing, weatherstripping, and hardware. All windows shall bear the NFRC (National Fenestration Rating Council) label indicating U-Factor and Solar Heat Gain Coefficient (SHGC) ratings.

B. Frame Material:
   1. Solid Vinyl Frames Only — Aluminum frames are prohibited due to thermal and condensation concerns.

C. Glazing:

1. Standard: All windows shall have, at a minimum, double-pane insulating glass units with a Low-(low-emissivity) coating and argon gas fill, Maximum U-factor 0.30, Maximum SHGC 0.25.
2. Safety Glazing: Windows in hazardous locations (e.g., adjacent to doors, in bathrooms, near stairs) shall have tempered safety glass as required by building code.

D. Hardware:
1. All operating hardware (locks, cranks, sash lifts) shall be factory-installed and designed for smooth, durable operation.
2. Finish: Hardware finish to be specified in the SOW (e.g., "White," "Satin Nickel").

E. Screens:
1. All operable windows shall be provided with a factory-supplied insect screen with an aluminum frame and fiberglass or aluminum screen mesh.

2.02 OPERATION

A. All new windows shall be horizontal sliders, single-slider configuration:
1. One operable sash.
2. One fixed sash.

2.03 INSTALLATION MATERIALS

A. Flashing Tape: Self-adhering, flexible flashing tape (butyl or modified bitumen based) designed for sealing window flanges to the sheathing.

B. Exterior Perimeter Sealant: High-performance, exterior-grade sealant conforming to ASTM C920, Class 25, Grade NS. Material shall be compatible with both the window frame and the adjacent siding/trim materials.

C. Foam Sealant: Low-pressure, low-expansion polyurethane foam sealant specifically formulated for insulating the gap between the window frame and the rough opening without bowing or deforming the frame.

2.04 WINDOW COUNTS

A. As clarified in RFI responses:
1. Crestview Cabin 1119: 10 windows.
2. Crestview Cabin 1120: 8 windows.
3. Crestview Cabin 1122: 8 windows.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Verify Openings: Before beginning installation, verify that rough openings are plumb, level, square, and correctly dimensioned for the window units to be installed.

B. Substrate Condition: Inspect the condition of the sheathing, framing, and weather-resistive barrier (WRB) around the opening. Report any rot, damage, or other unsatisfactory conditions to the COR in writing before proceeding.

3.02 PREPARATION

A. Clean Surfaces: Ensure the rough opening and surrounding sheathing are clean, dry, and free of any debris or projections that could interfere with the installation or damage flashing materials.

B. Prepare WRB: Prepare the weather-resistive barrier at the opening by cutting it in a modified "I" pattern or as recommended by the window manufacturer, allowing the flaps to be temporarily folded into the opening.

3.03 INSTALLATION

A. General: Install all window units and installation materials in strict accordance with the manufacturer's written instructions and all applicable building codes. The installation must create a continuous, watertight drainage plane.

B. Flashing and Sealing Sequence:
   1. Sill Flashing: Apply a continuous piece of self-adhering flashing tape to the sill of the rough opening, extending up the jambs a minimum of 6 inches on each side.
   2. Sealant: Apply a continuous bead of sealant to the back of the window's nailing flange on the top and sides, as recommended by the manufacturer. Do not apply sealant to the bottom flange to allow for drainage.
   3. Set Window: Set the window unit in the opening, ensuring it is plumb, level, and square. Secure the unit by fastening through the nailing flange into the wall framing as per the manufacturer's nailing schedule.
   4. Jamb Flashing: Apply flashing tape over the side (jamb) nailing flanges, extending a minimum of 5 inches above the top of the window and 5 inches below the bottom of the window.
   5. Head Flashing: Apply flashing tape over the top (head) nailing flange, extending a minimum of 5 inches beyond the side flashing on each side.
   6. Integrate WRB: Fold the flaps of the weather-resistive barrier back over the side and head flashing tapes and tape the seams. This ensures that any water getting behind the siding is directed over the top of the window, not into the wall cavity.
   7. Trim: Provide complete trim restoration.

C. Interior Seal:
   1. From the interior, apply low-expansion foam sealant into the gap between the window frame and the rough opening. Apply a consistent bead to create a continuous air seal without overfilling the cavity.

3.04 ADJUSTMENT AND CLEANING

A. Operation: After installation, operate all sashes and hardware to verify smooth and correct function. Adjust as necessary.

B. Cleaning: Remove all labels, protective films, and construction debris from the window glass and frames. Clean all surfaces in accordance with the manufacturer's recommendations, leaving them free of streaks, fingerprints, and construction marks.

3.05 PROTECTION

A. Protect all installed windows from damage by subsequent construction activities. The Contractor is responsible for repairing or replacing any window that is damaged before final project acceptance.

END OF SECTION

SECTION 08 70 00

HARDWARE

PART 1 - GENERAL

1.01 SUMMARY

A. Covers locksets, latches, closers, hinges, door stops, weatherstripping.

PART 2 - PRODUCTS

A. Exterior locksets: ANSI/BHMA Grade 2 minimum.
B. Interior locksets: Privacy or passage function as appropriate.
C. Finishes must match throughout each building: Satin Nickel or Government-approved equal.

PART 3 - EXECUTION

A. Install hardware per manufacturer instructions.
B. Adjust for smooth, quiet operation.

END OF SECTION

SECTION 08 90 00

LOUVERS AND VENTS

PART 1 - GENERAL

1.01 SUMMARY

A. Applies to installation of a metal airflow grate at Smokey Club House, Bedroom #1.

PART 2 - PRODUCTS

A. Grille shall be:
   1. Heavy-duty metal 18" x 18".
   2. Fixed louver design.
   3. Painted or powder-coated finish.
   4. Optional insect screen behind grate.

PART 3 - EXECUTION

A. Cut existing door to accept grate.
B. Install grille centered and secure.
C. Smooth and seal all cut edges.
D. Confirm grate provides adequate airflow for heater intake.

END OF SECTION

DIVISION 09 - FINISHES

SECTION 09 21 16

GYPSUM BOARD ASSEMBLIES

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for gypsum board (drywall) installation, finishing, and repair.

B. Includes installation of gypsum board for:
   1. Bathroom wall repairs.
   2. Ceiling tile replacement areas converted to drywall.
   3. Substrate repair after removal of fixtures and finishes.

C. Incorporates RFI clarifications regarding moisture-resistant board, Level 4 finish, and required backing/support.

1.02 RELATED REQUIREMENTS

A. Division 06 - Wood, Plastics, and Composites
B. Division 09 - Finishes.
C. Division 07 - Thermal and Moisture Protection.

1.03 REFERENCES

A. Gypsum Association (GA) - GA-216: Application and Finishing of Gypsum Panel Products.

B. ASTM International (ASTM) C1396 - Standard Specification for Gypsum Board.

1.04 SUBMITTALS

A. Product Data: For each type of gypsum board, finishing compound, and fastener.

PART 2 - PRODUCTS

2.01 GYPSUM BOARD PANELS

A. General Purpose Gypsum Board:
   1. Shall conform to ASTM C1396.
   2. Thickness: 1/2-inch for standard applications. Use 5/8-inch "Type X" for fire-rated assemblies or where specified for increased rigidity and sound control.

B. 5/8” Moisture- and Mold-Resistant Gypsum Board:
   1. Shall conform to ASTM C1396, with a moisture-resistant core and treated paper or fiberglass facings to resist mold growth.

2. For use in all bathrooms, laundry areas, and other areas subject to high humidity.

2.02 FINISHING MATERIALS

A. Joint Compound:
   1. Shall be all-purpose, drying-type joint compound conforming to ASTM C475.
   2. Alternatively, a system of separate taping and topping compounds may be used.

B. Joint Tape:
   1. Shall be paper joint tape for all flat joints and inside corners.
   2. Fiberglass mesh tape may be used only if embedded in a setting-type compound, not a drying-type compound.

2.03 FASTENERS AND TRIM

A. Fasteners:
   1. Shall be drywall screws conforming to ASTM C1002.
   2. Length shall be sufficient to penetrate wood framing members by at least 5/8 inch.

B. Trim Accessories:
   1. Shall be galvanized steel, vinyl, or paper-faced metal corner beads and trim accessories for all outside corners, edges, and terminations.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Framing Inspection: Before beginning installation, verify that all framing members are properly aligned, plumb, and level. Ensure that the spacing of framing members does not exceed the maximum span recommended for the specified gypsum board thickness.

B. Substrate Condition: Report any conditions that may adversely affect the quality of the installation (e.g., bowed studs, misaligned framing) to the COR in writing. Do not proceed with installation until unsatisfactory conditions have been corrected.

3.02 INSTALLATION (HANGING)

A. General: Install all gypsum board in strict accordance with Gypsum Association standard GA-216 and the manufacturer's written instructions.

B. Provide solid blocking behind all fixtures and wall-hung equipment.

C. Layout: Install panels with the long dimension horizontal to minimize the number of vertical joints. Stagger all end joints in successive courses.

D. Fastening:
   1. Fasten gypsum board to wood framing with the specified drywall screws.
   2. Space fasteners a maximum of 12 inches on center for ceilings and 16 inches on center for walls.
   3. Drive fasteners so that the head is set slightly below the surface of the board (dimpled) without breaking the paper face.

3.03 FINISHING

A. Level of Finish: Unless otherwise specified in the Statement of Work (SOW), all gypsum board surfaces shall receive a Level 4 Finish as defined by the Gypsum Association.

B. Level 4 Finish Procedure:
   1. First Coat: Embed joint tape in joint compound, covering all joints, angles, and trim accessories. Spot all fastener heads.
   2. Second Coat: After the first coat is dry, apply a second coat of joint compound over all taped joints and fastener heads. Feather the edges smooth.
   3. Third Coat: After the second coat is dry, apply a thin finish coat of joint compound over all joints and fastener heads.
   4. Sanding: After the final coat is thoroughly dry, lightly sand all finished surfaces to be smooth, uniform, and free of tool marks, ridges, or depressions.
   5. Ceilings must be smooth finish — no texture allowed.
   6. Walls may receive light orange-peel texture where existing finishes dictate.

C. Corner Beads: Install metal or vinyl corner beads on all outside corners to provide a durable, sharp edge.

3.04 REPAIR

A. Patching: Repair all holes, cracks, and other damage in existing gypsum board surfaces as specified in the SOW.

B. Method: For larger holes, cut the damaged area back to sound material in a square or rectangular shape. Install a new piece of gypsum board as a patch, securely fastened. Tape and finish the patch to blend seamlessly with the surrounding wall surface.

C. Final Appearance: All repaired areas shall be indistinguishable from the adjacent, original surface after final painting.

3.05 CEILING TILE TO DRYWALL CONVERSIONS

A. Remove entire suspension grid.
B. Install new furring or backing to create a level substrate.
C. Install 5/8" Type X board on ceilings.

3.06 FIELD QUALITY CONTROL

A. Inspect finished surfaces under normal lighting conditions. Surfaces shall be smooth and free of defects visible from 5 feet.

B. The COR may use oblique lighting to check for imperfections if a higher level of finish (Level 5) is specified.

3.07 CLEANUP AND PROTECTION

A. Keep the work area clean daily, controlling the spread of dust.

B. Upon completion of sanding, remove all dust from surfaces before priming and painting.

C. Protect installed gypsum board from moisture and physical damage.

END OF SECTION

SECTION 09 51 00

ACOUSTICAL CEILINGS

PART 1 - GENERAL

1.01 SUMMARY

A. Applies to in-kind ceiling tile replacement at buildings where tile ceilings remain.
B. Includes complete field replacement for uniform appearance.

PART 2 - PRODUCTS

A. Commercial-grade mineral fiber tiles by Armstrong, USG, or CertainTeed.

B. Minimum performance:
   1. Class A fire rating (ASTM E84).
   2. NRC $\geq$ 0.55.

C. Tile size:

   1. Standard 2x2 or 2x4 tiles.
   2. Replace entire room if existing size is obsolete.

D. Edge profile: Square lay-in.

PART 3 - EXECUTION

A. Remove all existing mismatched or aged tiles.
B. Repair and level suspension grid.
C. Install full matching set for each enclosed ceiling area.

END OF SECTION

SECTION 09 60 00

FLOORING

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for the removal of existing flooring and the installation of new flooring systems, including underlayment, adhesives, and accessories.

1.02 SUBMITTALS

A. Product Data: For each type of flooring, underlayment, and adhesive.

B. Samples: Submit samples of each specified flooring material, showing the full range of available colors and patterns for selection by the COR.

C. Warranty: Submit sample copies of the manufacturer's material warranty.

1.03 DELIVERY, STORAGE, HANDLING

A. Deliver flooring materials to the site in their original, unopened packaging. Store materials in a dry, climate-controlled area, and allow them to acclimate to the installation environment for the period recommended by the manufacturer (typically 48 hours).

PART 2 - PRODUCTS

2.01 Luxury Vinyl Plank (LVP):

A. Material: Shall be a rigid core, 3.0 mm minimum thickness waterproof LVP with an attached underlayment pad.
B. Wear Layer: Minimum 20 mil wear layer for commercial-grade durability.
C. Installation Type: Glue-down installation only — floating floors prohibited
D. Acceptable lines: Mannington Commercial, Armstrong, or equal.
E. Applies to kitchens, living rooms, hallways, and common areas (not bathrooms).

2.02 CARPET

A. Carpet Tile Only — broadloom carpet is prohibited.

B. Minimum performance:
   1. 20 oz/yd² face weight.
   2. Solution-dyed nylon fiber.
   3. Fiberglass-reinforced backing (e.g., Shaw EcoWorx).
   4. Applies to carpet replacement in living spaces and bedrooms in multiple barracks and houses.
   5. Direct glue-down with pressure-sensitive adhesive.
   6. No separate carpet pad permitted.

2.03 SHEET VINLY

A. Material: Shall be a rigid core, 2.0 mm minimum thickness waterproof sheet vinyl with an attached underlayment pad.
B. Wear Layer: Minimum 20 mil wear layer for commercial-grade durability.
C. Installation Type: Heat-welded seams in bathrooms.
D. Acceptable lines: Mannington Commercial, Armstrong, or equal.
E. Applies to sheet vinyl installation primarily in bathrooms and wet areas.

2.04 PORCELAIN TILE

1. Material: Conforming to ANSI A137.1, with a PEI (Porcelain Enamel Institute) wear rating of 4 or 5 for heavy traffic.
2. Finish: Slip-resistant finish suitable for the intended application (e.g., bathrooms, entryways).
3. Selection: Manufacturer, size, color, and pattern to be specified in the SOW.

2.05 INSTALLATION MATERIALS

A. Underlayment:
   1. Floor Patch/Leveling Compound: Portland cement-based compound for patching or leveling subfloors as needed.

B. Adhesives:
   1. All adhesives shall be recommended by the flooring manufacturer for the specific product and substrate. Adhesives shall be low-VOC.

C. Tile Setting Materials:
   1. Mortar: Polymer-modified thin-set mortar conforming to ANSI A118.4.
   2. Grout: Polymer-modified or epoxy grout, with a color selected by the Owner. Grout shall be stain resistant.

D. Transition Strips and Base:
   1. Transition Strips: Provide transition strips at all doorways and where new flooring meets a different flooring type. Material and finish to be specified in the SOW (e.g., "Aluminum, Satin Nickel finish").
   2. Wall Base: Provide new wall base in all areas with new flooring. Material to be specified in the SOW (e.g., "4-inch vinyl cove base" or "4-1/4-inch paint-grade wood base").

PART 3 - EXECUTION

3.01 PREPARATION

A. Removal of Existing Flooring: Completely remove all existing flooring, underlayment, tack strips, staples, and adhesives down to a clean, bare subfloor.

B. Subfloor Examination: Inspect the subfloor for any damage, moisture, or defects. Report any unsatisfactory conditions to the COR in writing. Do not proceed until the subfloor is sound.

C. Subfloor Preparation:
   1. Prepare the subfloor to be clean, smooth, dry, and structurally sound.

2. Scrape or grind any remaining adhesive residue.
3. Use the specified floor patch or leveling compound to fill any cracks, holes, or low spots. The final subfloor must be flat to within the tolerances required by the flooring manufacturer (typically 1/8 inch in 6 feet or 3/16 inch in 10 feet).
4. Vacuum the entire area immediately before beginning installation.

3.02 INSTALLATION

A. General: Install all flooring, underlayment, and accessories in strict accordance with the manufacturer's written instructions and industry best practices.

B. Luxury Vinyl Plank (LVP) Installation:
   1. Install flooring with staggered end joints for a random, natural appearance.
   2. Maintain the manufacturer's recommended expansion gap around the entire perimeter of the room. Do not install flooring tight against walls or other fixed objects.
   3. Install LVP using premium adhesive.
   4. Install new transitions and vinyl cove base throughout replaced areas.

C. Carpet Installation:
   1. Install carpet pad over the subfloor.
   2. Install new tack strips around the perimeter of the room.
   3. Power-stretch all carpet to ensure it is taut and to prevent wrinkling.
   4. Level substrate to eliminate telegraphing.
   5. Install carpet tile in straight-lay or quarter-turn pattern (as directed).
   6. Seam all carpet sections in inconspicuous locations where possible. All seams shall be made with a high-quality seam tape and a seaming iron, resulting in a strong, flat, and minimally visible seam.

D. Tile Installation:
   1. Install tile in accordance with the TCNA (Tile Council of North America) Handbook.
   2. Use a chalk line or laser to establish a square layout.
   3. Achieve full mortar coverage on the back of each tile to ensure a strong bond and prevent cracking.
   4. Maintain consistent grout joint widths.
   5. After the mortar has cured, grout all joints, ensuring they are full and free of voids. Clean all excess grout from the tile surfaces.

E. Wall Base and Transition Strips:
   1. Install new wall base after the flooring is installed, fitting it tightly to the floor and wall.
   2. Install transition strips at all doorways and where new flooring meets a different flooring type, providing a smooth and safe transition.

3.03 CLEANING AND PROTECTION

A. Keep the work area clean daily, removing all scrap materials and debris.

B. Upon completion of installation, vacuum and clean the new flooring in accordance with the manufacturer's recommendations.

C. Protect the new flooring from damage by subsequent construction activities. Use protective coverings like Ram Board or clean cardboard in all traffic areas until final project acceptance.

END OF SECTION

SECTION 09 90 00

PAINTING AND COATING

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for the preparation and painting of interior and exterior surfaces across all buildings.

B. Based on extensive RFI clarifications regarding:
   1. Measurement methodology
   2. Required sheens
   3. Surfaces included
   4. Lead-safe practices

1.02 RELATED REQUIREMENTS

A. Division 02 - Hazardous Materials Discovery Protocol.
B. Division 06 - Wood, Plastics, and Composites.
C. Division 07 - Thermal and Moisture Protection.

1.03 SUBMITTALS

A. Product Data: For each type of paint, primer, and sealant to be used.

B. Color Schedule: Submit a schedule of all surfaces to be painted, indicating the specified color and sheen for each.

1.04 QUALITY ASSURANCE

A. Mock-up: If requested by the COR, apply a sample of each color and sheen to a small, inconspicuous area for approval before proceeding with the full application.

PART 2 - PRODUCTS

2.01 PAINT

A. General: All paint shall be premium quality, acrylic latex. Manufacturer, product line, and colors to be specified in the Statement of Work (SOW).

B. Interior Paint: Shall be low-odor, low-VOC.
   1. Ceilings: Flat or matte finish.
   2. Walls (General Areas): Eggshell or satin finish.
   3. Walls (Bathrooms, Kitchens) and all Trim, Doors, and Cabinets: Semi-gloss or satin finish for durability and washability.
   4. Acceptable Brands: Sherwin-Williams (ProMar 200 HP, Duration, Emerald) or Benjamin Moore (Ultra Spec, Regal Select).

C. Exterior Paint: Shall be 100% acrylic latex, formulated for exterior use on the specified substrate (wood, fiber cement, stucco, etc.).
   1. Siding and Body: Flat or satin finish.
   2. Trim and Doors: Satin or semi-gloss finish.
   3. Three-coat system:
      a) Primer (manufacturer-recommended).
      b) Two finish coats.

D. Color Selection
   1. Colors to match existing unless directed otherwise by Government.
   2. No accent walls required.

2.02 PRIMERS

A. General: All primers shall be as recommended by the paint manufacturer for the specific substrate and conditions.

B. New Drywall Primer: PVA (polyvinyl acetate) drywall primer-sealer.

C. Bare Wood Primer: High-quality interior/exterior acrylic or oil-based wood primer with stain-blocking properties.

D. Stain-Blocking Primer: High-adhesion primer designed to block water stains, wood tannins, smoke, and other blemishes.

E. Metal Primer: Rust-inhibitive primer designed for ferrous metals (e.g., steel).

2.03 PREPARATION MATERIALS

A. Cleaners: Mild detergent, TSP (Trisodium Phosphate) substitute, or other appropriate cleaning agents.

B. Fillers:
   1. Interior: Sandable spackling compound for drywall; paintable wood filler for wood trim.
   2. Exterior: Exterior-grade, sandable wood filler or epoxy-based repair compound

C. Caulk/Sealant: Paintable, acrylic latex sealant conforming to ASTM C834.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Before beginning work, examine all substrates to be painted. Report any conditions that may adversely affect the quality of the finish (e.g., moisture, damage, or surface defects) to the COR in writing. Do not proceed until unsatisfactory conditions have been corrected.

B. Contractor must assume presence of Lead-Based Paint in pre-1978 buildings.

3.02 SURFACE PREPARATION

A. General: The quality of the final finish is directly dependent on the quality of the surface preparation. All surfaces shall be prepared in strict accordance with the paint manufacturer's instructions to ensure proper adhesion and a uniform finish.

B. Protection: Protect all adjacent surfaces, floors, furniture, hardware, and fixtures not being painted by covering them with drop cloths, plastic sheeting, and masking tape.

C. Cleaning: Clean all surfaces to be painted to remove any dirt, dust, grease, oil, and other contaminants. Wash surfaces with a suitable cleaner and allow them to dry completely.

D. Removal of Loose Paint: Scrape and sand all loose, peeling, cracking, or blistering paint down to a sound, feathered edge.

E. Patching and Filling:
   1. Fill all nail holes, cracks, and minor imperfections with the specified filler.
   2. After the filler is dry, sand all patched areas smooth and flush with the surrounding surface.
   3. Large repairs are covered under the siding/drywall line items.

F. Caulking: Remove all old, failed caulk. Apply a continuous bead of new, paintable caulk to fill any gaps between trim and adjacent walls or ceilings. Tool the bead for a smooth, concave finish.

G. Priming:
   1. Apply the appropriate primer to all bare wood, new drywall, patched areas, and any surfaces with stains (e.g., water stains, wood tannins) that could bleed through the topcoat.
   2. Allow primer to dry completely before applying the finish coat.

3.03 APPLICATION

A. General: Apply paint in accordance with the manufacturer's written instructions. Do not thin paint unless specifically permitted by the manufacturer.

B. Environmental Conditions: Apply paint only when the temperature of the air and the surface are within the range recommended by the manufacturer (typically between 50°F and 90°F).

C. Coats: Apply a minimum of two (2) finish coats to all surfaces to achieve a uniform, even color and sheen. A full prime coat may be required for significant color changes or over new surfaces, in addition to the two finish coats.

D. Workmanship: Apply paint to produce a smooth, uniform finish free of drips, runs, sags, brush marks, roller marks, or other imperfections. Cut in all edges neatly and cleanly.

3.04 FIELD QUALITY CONTROL

A. The final finish shall be inspected under normal lighting conditions. The finished surface shall be uniform in color, sheen, and texture.

B. The finish shall be free of any visible defects from 5 feet.

3.05 CLEANUP AND PROTECTION

A. Provide temporary dust partitions and protect Government personnel.

B. Move furniture and appliances as required; protect stored items.

C. Provide HEPA vacuuming, containment, and cleanup.

D. Keep the work area clean daily.

E. Upon completion of painting in each area, remove all masking materials and protective coverings. Clean up all spills and drips.

F. Touch up any damage to new paintwork before final project acceptance.

G. Properly dispose of all waste materials, including empty paint cans and rags, in accordance with local environmental regulations.

END OF SECTION

DIVISION 10 - SPECIALTIES

SECTION 10 28 00

TOILET, BATH AND LAUNDRY ACCESSORIES

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for the installation of common accessories such as:
   1. Shower rods.
   2. Grab bars (if required).
   3. Towel bars.
   4. Bathroom hardware and other wall-mounted items.

B. Incorporates RFI clarifications requiring proper wall blocking behind all wall-mounted accessories.

1.02 RELATED REQUIREMENTS

A. Division 06 - Blocking requirements for wall-hung elements.
B. Division 09 - Drywall and paint finish requirements.
C. Division 22 - Plumbing fixture installations.

1.03 SUBMITTALS

A. Product Data: For each type of accessory, showing material, finish, and dimensions.

1.04 COORDINATION

A. Coordinate with the work of Section 06 10 00 - Rough Carpentry to ensure that solid wood blocking is installed in walls for the secure attachment of all accessories, especially grab bars.

PART 2 - PRODUCTS

2.01 MATERIALS AND FINISH

A. General: All accessories within a single room shall be from a single manufacturer to ensure a consistent style and finish.

B. Material: Materials shall be stainless steel or heavy-duty composite designed for housing/barracks use.

C. Finish: Finishes shall match door hardware and plumbing trim (Satin Nickel unless otherwise approved).

2.02 ACCESSORIES

A. Vanity and Bath Accessories:
   1. Tower bars, toilet paper holders, robe hooks, etc. shall match room finish.
   2. Commercial grade fastening hardware required.

B. Grab Bars:
   1. Standard: Shall comply with all applicable ADA (Americans with Disabilities Act) requirements for diameter (1-1/4 to 1-1/2 inches), wall clearance, and structural strength (capable of supporting a minimum of 250 pounds).
   2. Material: 18-gauge stainless steel.

C. Shower Rods:
   1. Type: Heavy-duty stainless-steel curved or straight rods.
   2. Secure, screw-mounted brackets only (no tension rods permitted).

PART 3 - EXECUTION

3.01 EXAMINATION

A. Blocking Verification: Before drywall is installed, verify that solid wood blocking has been installed in the wall framing at all specified locations for accessory installation, particularly for grab bars.

B. Surface Condition: After finishes are complete, examine wall surfaces to ensure they are clean, dry, and ready for accessory installation.

3.02 INSTALLATION

A. General: Install all accessories in strict accordance with the manufacturer's written instructions.

B. Location: Install accessories at the locations and heights specified in the Statement of Work (SOW) or as directed by the COR. If not specified, install at standard, conventional heights.

C. Anchoring:
   1. Secure Anchoring: All accessories shall be securely anchored to the wall.
   2. Grab Bars: Grab bars must be anchored directly into solid wood blocking or wall studs using the stainless-steel screws provided by the manufacturer. Do not use toggle bolts, hollow-wall anchors, or other non-structural fasteners for grab bar installation.
   3. Other Accessories: For accessories other than grab bars, where blocking is not available, use appropriate heavy-duty hollow-wall anchors (e.g., toggle bolts, snap-toggles) that are suitable for the substrate and the anticipated load.

D. Final Installation: Ensure all accessories are installed level, plumb, and rigid, with no looseness or play. All mounting flanges and fasteners shall be concealed by the accessory's cover plate (escutcheon).

3.03 FIELD QUALITY CONTROL

A. After installation, test all accessories to ensure they are securely attached to the wall.

B. Apply a firm load to all grab bars to verify their stability and structural integrity.

3.04 CLEANING

A. Upon completion, clean all accessories of any dust, fingerprints, or construction marks.

B. Leave accessories in a clean, polished, and ready-to-use condition.

END OF SECTION

SECTION 10 29 00

PEST CONTROL

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for the installation of physical barriers to prevent the entry of rodents and other pests into building structures.

B. RFI clarifications define installation depth, mesh gauge, burial geometry, and protected building areas.

1.02 RELATED REQUIREMENTS

Coordinate the installation of pest exclusion materials with work of other sections, to ensure a complete and integrated barrier.

A. Division 01 - General Requirements (site protection).
B. Division 07 - Exterior Work including siding and vent areas.
C. Division 31 - Earthwork (minor soil excavation for mesh burial).

PART 2 - PRODUCTS

2.01 EXCLUSION MATERIALS

A. Rodent-Proof Mesh:
   1. Material: Shall be galvanized steel hardware cloth or stainless-steel mesh. Copper mesh may be used for filling small gaps.
   2. Grid Size: Maximum 1/4-inch grid opening to effectively block mice and other small rodents. Using 1/2-inch openings is prohibited.
   3. Corrosion-resistant fasteners required.
   4. Shall include fastening washers and screws for attachments to structural surfaces.

B. Sealants:
   1. Pest-Resistant Sealant: Shall be a heavy-duty elastomeric or polyurethane sealant suitable for exterior use. Shall be used on gaps <1/4 inch.
   2. Filler Material: For larger gaps, use sealant in conjunction with a filler material like copper mesh or coarse steel wool to prevent rodents from chewing through.

C. Vent Screens:
   1. Material: Shall be heavy-duty, corrosion-resistant metal screens designed for installation on exterior vents (e.g., foundation vents, exhaust vents).
   2. Shall include fastening washers and screws for attachments to structural surfaces.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Thorough Inspection: Before beginning installation, conduct a thorough visual inspection of the entire building exterior, from the foundation to the roofline, to identify all potential pest entry points.

B. Common Entry Points: Pay close attention to the following common entry points:
   1. Gaps at the base of siding where it meets the foundation.
   2. Cracks or holes in the foundation wall.
   3. Gaps around all utility penetrations, including pipes, conduits, and HVAC lines.
   4. Unscreened or damaged foundation, soffit, and exhaust vents.
   5. Gaps at outside corners and under eaves.

3.02 INSTALLATION

A. General: Install all pest exclusion materials to create a continuous, durable barrier that completely seals all identified and potential entry points.

B. Rodent-Proof Mesh Installation:
   1. Skirting/Barrier: Install rodent-proof mesh along the base of the building as specified in the SOW. The mesh shall be securely fastened to both the siding and the foundation wall to eliminate any gaps.
   2. Fastening: Use corrosion-resistant screws and washers or other appropriate mechanical fasteners spaced a maximum of 12 inches on center to secure the mesh.
   3. Burial: Where specified, bend the bottom edge of the mesh into an "L" shape, bury it at least 6 inches below grade and 6 inches away from the structure to prevent rodents from digging underneath.

C. Sealing Gaps and Cracks:
   1. For small cracks and gaps (less than 1/4 inch), apply a continuous bead of pest-resistant sealant to completely fill the opening.
   2. For larger gaps and holes, pack the opening tightly with copper mesh or coarse steel wool, then seal over the filler material with pest-resistant sealant.

D. Vent Screen Installation:
   1. Securely fasten heavy-duty metal screens over all exterior vent openings.
   2. Ensure screens are installed in a manner that does not significantly impede airflow but completely blocks pest entry.

3.03 LOCATION OF MESH INSTALLATION

A. Building base perimeter:
   1. Install mesh continuously along bottom edge of siding and foundation intersection.

B. Ventilation protection:
   1. Install mesh over crawlspace, foundation, and low exhaust vents without restricting airflow.

C. Penetrations:
   1. Install mesh tightly around exterior utility penetrations.

3.04 FIELD QUALITY CONTROL

A. Contractor must field-verify required linear footage before ordering materials.

B. Work to follow building perimeter quantities identified in the SOW.

C. Upon completion, conduct a final visual inspection of the entire building to verify that all identified entry points have been sealed.

D. Check that all installed mesh and screens are secure, taut, and free of any gaps at the edges.

3.05 CLEANUP

A. Keep the work area clean daily.

B. Upon completion, remove all scrap materials, fasteners, and debris from the site.

END OF SECTION

DIVISION 22 - PLUMBING

SECTION 22 40 00

PLUMBING FIXTURES

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for new plumbing fixtures, including kitchen sinks & faucets, bathroom sinks, vanities & faucets, toilets, shower heads & shower trim, and full shower valve bodies.

B. Includes labor, materials, equipment, and installation for all plumbing fixtures and systems identified in the Schedule of Work.

C. Incorporates all RFI clarifications regarding:
   1. Fixture material and performance requirements.
   2. Water and sewer connections at trailer pads.
   3. Replacement scope boundaries (e.g., shower fixtures, sinks, toilets).
   4. Exclusions (mold testing, environmental testing, extensive abatement).

1.02 RELATED REQUIREMENTS

A. Division 01 – General Requirements.
B. Division 03 – Concrete (for trailer pad trenches).
C. Division 10 – Bath Accessories.
D. Division 26 – Electrical (for water heaters and circuits).

1.03 SUBMITTALS

A. Product Data: For each type of fixture, showing materials, finishes, flow rates, and installation requirements.

1.04 QUALITY ASSURANCE

A. Regulatory Requirements: All fixtures shall comply with the EPA's WaterSense program requirements for water efficiency.

B. All rough-in components must match existing supply and waste connections.

C. All installations must comply with California Plumbing Code.

D. Installer Qualifications: All plumbing work shall be performed by a licensed plumber.

PART 2 - PRODUCTS

2.01 TOILETS

A. General: Toilets shall be two-piece, vitreous china, with an elongated bowl and a standard or "comfort" height (ADA-compliant).

B. Performance: All toilets shall be EPA WaterSense certified, with a maximum flush volume of 1.28 gallons per flush (GPF).

C. Components: Shall be provided as a complete unit, including tank, bowl, slow-close solid plastic toilet seat, wax ring, floor bolts, new stainless-steel supply line and new quarter-turn shutoff valve.

D. If flange or subfloor is damaged, notify COR; additional repairs handled as modifications.

2.02 LAVATORIES (BATHROOM SINKS)

A. Material: Vitreous china.

B. Type: Undermount or drop-in, as specified in the SOW to fit new countertops. All lavatories shall have an overflow drain.

C. Selection: Manufacturer, model, and size to be specified in the SOW.

2.03 KITCHEN SINKS

A. Material: 18-gauge stainless steel.

B. Configuration: Single-bowl or double-bowl, as specified in the SOW.

C. Features: Shall include sound-dampening pads on the underside.

D. Components: Shall include basket strainer drain assemblies.

2.04 FAUCETS

A. General: All faucets shall be lead-free compliant, and EPA WaterSense certified. Includes new supply lines and shutoff valves. Include new supply stops and braided stainless-steel connectors.

B. Lavatory Faucets:
   1. Material: Brass body with a durable finish (chrome or brushed nickel).
   2. Type: Single-handle or two-handle, as specified in the SOW.
   3. Flow Rate: Maximum 1.5 gallons per minute (GPM).

C. Kitchen Faucets:
   1. Material: Brass body with a durable finish (chrome or brushed nickel).
   2. Type: Single handle with a pull-out or pull-down sprayer.
   3. Flow Rate: Maximum 1.8 GPM.

2.05 SHOWER VALVES AND TRIM

A. Replace entire shower control assembly, including:
   1. Rough-in valve body.
   2. Trim plate.

B. Shower Valve: Shall be a pressure-balancing or thermostatic mixing valve to prevent scalding.

C. Shower Head: Shall be EPA WaterSense certified, with a maximum flow rate of 1.8 GPM.

D. Finish: To match other fixtures in the bathroom.

2.06 VANITIES (COORDINATED WITH DIVISION 06)

A. Vanities must match dimensional footprint of existing units.
B. New solid-surface or cultured marble countertops with integral bowls.
C. Wall-hung vanities require concealed carriers and new blocking.

2.07 INSTALLATION REQUIREMENTS

A. Supply Lines: Braided stainless steel.

B. Shut-off Valves (Stops): Chrome-plated brass, quarter-turn ball valve type.

C. P-Traps and Drains: PVC, ABS, or chrome-plated brass as required by local code and application.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Verify Rough-In: Before beginning installation, verify that all rough-in water supply and drain piping are correctly located, properly sized, and capped.

B. Inspect Substrates: Examine all substrates (floors, countertops, walls) to ensure they are sound, level, and ready to receive new fixtures. Report any unsatisfactory conditions to the COR in writing.

3.02 INSTALLATION

A. General:
   1. Install all plumbing fixtures, trim, and accessories in strict accordance with the manufacturer's written instructions and all applicable state and local plumbing codes.
   2. Install fixtures level, secure, and watertight.
   3. Replace all supply lines, stops, traps, escutcheons.
   4. Test all fixtures for leaks.

B. Toilets:
   1. Set toilets on a new wax ring to create a positive seal with the floor flange.
   2. Securely fasten the toilet to the floor with new floor bolts. Do not over-tighten bolts to avoid cracking the porcelain.
   3. Install bolt caps.

4. Shim for level as required.
5. Confirm proper flush operation.
6. Seal base with silicone after testing.

C. Lavatories and Sinks:
1. Securely mount sinks to countertops in accordance with the manufacturer's instructions.
2. Apply a continuous bead of 100% silicone sealant around the perimeter of drop-in sinks to create a watertight seal.

D. Faucets and Drains:
1. Install faucets, drain assemblies, and P-traps, ensuring all connections are tight.

E. Shower Installation

A. Install new shower valve body at correct rough-in depth.
B. Test valve operation before wall closure.
C. Install new shower head and trim kit.

F. Supply Connections:
1. Install new quarter-turn shut-off valves (stops) on all supply lines.
2. Connect fixtures to the shut-off valves using new, braided stainless-steel supply lines.

3.03 TESTING

A. Water Test: After installation is complete, turn on the water supply and fully test all fixtures.
1. Operate all faucets and shower valves at full pressure.
2. Flush all toilets multiple times.
3. Fill each sink bowl and then drain it.

B. Leak Inspection: While testing, thoroughly inspect all connections, including supply lines, shut-off valves, drain assemblies, and toilet seals, for any signs of leaks.

C. Correction: All leaks shall be immediately repaired and re-tested until the system is proven to be 100% watertight.

3.04 ADJUSTMENT AND CLEANING

A. Adjust faucet handles and toilet flushing mechanisms for proper operation.

B. Upon completion, clean all fixtures of any dust, sealant residue, and construction marks. Polish chrome and other finished surfaces.

C. Remove all labels from new fixtures.

3.05 PROTECTION

A. Protect all installed fixtures from damage by subsequent construction activities until final project acceptance.

END OF SECTION

SECTION 22 11 00

FACILITY WATER DISTRIBUTION

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for underground exterior water distribution piping, valves, and hydrants for service to trailer pads and other site locations.

1.02 SUBMITTALS

A. Product Data: For each type of pipe, valve, fitting, and hydrant.

B. Test Reports: Submit reports for all required pressure tests and water quality tests.

1.03 QUALITY ASSURANCE

A. Regulatory Requirements: All work shall comply with local health department regulations and plumbing codes.

B. Installer Qualifications: All plumbing work shall be performed by a licensed plumber experienced in site utility installation.

PART 2 - PRODUCTS

2.01 UNDERGROUND WATER PIPING

A. Pipe Material: All pipe and fittings shall be rated for potable water use (NSF/ANSI 61) and have a minimum pressure rating of 160 psi. Options include:
   1. Cross-linked Polyethylene (PEX): Conforming to ASTM F876. Type PEX-A is preferred for its flexibility and freeze-resistance.
   2. Copper Tube: Type K, soft temper, conforming to ASTM B88.

2.02 VALVES AND FITTINGS

A. General: All valves and fittings shall be lead-free compliant.

B. Fittings: Shall be compatible with the selected pipe material.
   1. For PEX: Brass or high-performance polymer fittings (e.g., ASTM F1960 expansion type).
   2. For Copper: Wrought copper fittings.

C. Valves:
   1. Shut-off Valves: Shall be heavy-duty, lead-free brass ball valves, rated for underground service.

2.03 HYDRANTS/HOSE BIBS

A. Yard Hydrants:
   1. Type: Shall be a frost-proof yard hydrant, designed to drain water from the riser pipe when shut off to prevent freezing.
   2. Material: Heavy-duty cast iron head and handle with a brass or bronze valve body.
   3. Burial Depth: Shall have a burial depth sufficient to place the valve body below the local frost line.
   4. Backflow Prevention: Shall be equipped with an integral vacuum breaker to prevent back-siphonage.

2.04 **WARNING TAPE**

A. Material: Shall be a 4- to 6-inch wide, acid- and alkali-resistant polyethylene tape.

B. Color and Text: Tape shall be BLUE and imprinted with "CAUTION: WATER LINE BELOW" or similar wording.

PART 3 - EXECUTION

3.01 **EXAMINATION**

A. Utility Location: Before beginning any trenching or excavation, contact the local utility locating service ("Call Before You Dig") and verify the locations of all existing underground utilities, including gas, electricity, and communications lines.

B. Contractor’s scope starts at existing main distribution line or isolation valve box.

C. Site Conditions:
   1. Examine the proposed route for the new water line and report any obstructions or conditions that may affect the installation to the COR.
   2. Contractor must field-verify water pressure and valve condition.

3.02 **TRENCHING**

A. Depth: Excavate trenches to a depth that provides a minimum cover of 12 inches over the top of the pipe, or a depth sufficient to place the pipe below the local frost line, whichever is deeper.

B. Trench Bottom: The bottom of the trench shall be stable, uniform, and free of sharp rocks or debris that could damage the pipe.

C. Bedding: Place a minimum of 4 inches of sand or fine gravel bedding in the trench bottom to provide continuous support for the pipe.

3.03 **INSTALLATION**

A. General:
   1. Install all piping, valves, and hydrants in strict accordance with the manufacturer's written instructions and all applicable plumbing codes.
   2. Maintain continuous positive slope to prevent freezing.

B. Pipe Laying: Lay pipe on the prepared bedding. Do not lay pipe in standing water.

C. Joints and Connections: Make all joints and connections watertight and in accordance with the manufacturer's recommendations for the specific pipe and fitting system being used.

D. Hydrant Installation:
   1. Set frost-proof yard hydrants plumb and at the locations specified in the SOW.
   2. Create a drainage pit at the base of each hydrant by placing a minimum of 2 cubic feet of coarse gravel or crushed stone around the drain hole. This is critical to allow the hydrant to self-drain and prevent freezing.

3.04 BACKFILLING

A. Initial Backfill: After the pipe has been tested and approved, carefully place and compact an initial layer of sand or fine gravel backfill to a depth of 6 inches over the top of the pipe to protect it from damage.

B. Warning Tape: Place the specified blue warning tape in the trench, approximately 12 inches above the top of the pipe.

C. Final Backfill: Complete the backfilling of the trench with the excavated material, compacting it in lifts to prevent future settlement. Crown the trench slightly to allow for minor settlement.

D. Surface Restoration: Restore any disturbed surface areas (e.g., lawn, pavement) to their original condition.

3.05 TESTING AND DISINFECTION

A. Pressure Testing:
   1. After installation but before backfilling, hydrostatically pressure test the entire new water line system.
   2. The test pressure shall be a minimum of 150 psi or 1.5 times the system working pressure, whichever is greater, for a minimum duration of 2 hours, with no loss of pressure.

B. Disinfection (Chlorination):
   1. After a successful pressure test, thoroughly flush the new water lines to remove any debris.
   2. Disinfect the entire system in accordance with AWWA Standard C651, typically by introducing a chlorine solution and allowing it to stand for 24 hours.

C. Final Flush and Water Quality Test:
   1. After the disinfection period, thoroughly flush the system again until the chlorine residual matches that of the main water supply.
   2. Collect water samples and submit them to a certified laboratory for bacteriological testing (testing for coliform bacteria).
   3. The system shall not be placed into service until a satisfactory water quality report has been received and submitted to the COR.

3.06 CLEANUP

A. Upon completion, remove all excess materials, tools, and debris from the site. Leave the site in a clean and restored condition.

END OF SECTION

SECTION 22 13 00

SANITARY SEWER SYSTEMS

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for underground exterior sanitary sewer piping, cleanouts, and connections for service to trailer pads and other site locations.

1.02 SUBMITTALS

A. Product Data: For each type of pipe, fitting, and cleanout.

1.03 QUALITY ASSURANCE

A. Regulatory Requirements: All work shall comply with local health department regulations and plumbing codes.

B. Installer Qualifications: All plumbing work shall be performed by a licensed plumber experienced in site utility installation.

PART 2 - PRODUCTS

2.01 UNDERGROUND SANITARY SEWER PIPING

A. Pipe Material: Shall be Polyvinyl Chloride (PVC) pipe, conforming to ASTM D3034, SDR 35. All pipe and fittings shall be designed for gravity sewer applications.

B. Fittings: Shall be PVC, compatible with the pipe, with watertight gasketed or solvent-weld joints.

2.02 CLEANOUTS

A. General: Cleanouts shall consist of a fitting on the main sewer line, a riser pipe, and a threaded plug or cap at finished grade.

B. Materials: All underground fittings and riser pipe shall be PVC, same as the main sewer line.

C. Plugs and Covers:
   1. In Landscape Areas: Heavy-duty, threaded PVC plug.
   2. In Traffic Areas (driveways, pads): Heavy-duty, cast-iron frame and cover marked "SEWER".

2.03 WARNING TAPE

A. Material: Shall be a 4- to 6-inch wide, acid- and alkali-resistant polyethylene tape.

B. Color and Text: Tape shall be GREEN and imprinted with "CAUTION: SEWER LINE BELOW" or similar wording.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Utility Location: Before beginning any trenching or excavation, contact the local utility locating service ("Call Before You Dig") and verify the locations of all existing underground utilities.

B. Verify Elevations: Verify the invert elevations of the connection points (e.g., at the trailer pad and the main sewer line) to ensure that the planned pipe run can achieve a continuous, positive slope. Report any discrepancies to the COR before proceeding.

C. Unknown Depths**:** Contractor must pothole or hand-dig to locate existing sewer line if depths are not known.

3.02 TRENCHING

A. Depth: Excavate trenches to a depth sufficient to install the pipe at the required slope while maintaining a minimum cover of 12 inches or placing it below the local frost line, whichever is deeper.

B. Trench Bottom: The bottom of the trench shall be stable, uniform, and excavated to a continuous grade. It must be free of sharp rocks or debris that could damage the pipe.

C. Bedding: Place a minimum of 4 inches of sand or fine gravel bedding in the trench bottom, graded to provide continuous support for the pipe at the specified slope.

3.03 INSTALLATION

A. General: Install all piping, fittings, and cleanouts in strict accordance with the manufacturer's written instructions and all applicable plumbing codes.

B. Connection Points: Contractor scope begins at existing main sewer line lateral identified in field.

C. Slope:
1. Install all sewer piping at a uniform slope to ensure gravity flow.
2. Unless otherwise specified, the minimum slope shall be 1/4 inch per foot (2% grade).

D. Pipe Laying:
1. Lay pipe on the prepared bedding, starting from the lowest point and working upgrade.
2. Ensure the bell ends of the pipe face upgrade.

E. Joints and Connections:
1. Make all joints watertight. For solvent-weld joints, use the appropriate primer and cement. For gasketed joints, ensure the gasket and bell are clean before insertion.

F. Cleanouts:
   1. Install cleanouts at locations specified in the SOW, including the upper end of each line, at changes in direction greater than 45 degrees, and at intervals not exceeding 100 feet for straight runs.
   2. Extend cleanout risers vertically to the finished grade and install the appropriate plug or cover.

3.04 BACKFILLING

A. Initial Backfill: After the pipe has been tested and approved, carefully place and compact an initial layer of sand or fine gravel backfill to a depth of 6 inches over the top of the pipe to protect it from damage.

B. Warning Tape: Place the specified green warning tape in the trench, approximately 12 inches above the top of the pipe.

C. Final Backfill: Complete the backfilling of the trench with the excavated material, compacting it in lifts to prevent future settlement. Crown the trench slightly to allow for minor settlement.

D. Surface Restoration: Restore any disturbed surface areas (e.g., lawn, pavement) to their original condition.

3.05 TESTING

A. General: Test all new sewer lines for leaks before backfilling.

B. Test Method: Perform a low-pressure air test or a water exfiltration test in accordance with local plumbing code requirements.

C. Visual Inspection: Visually inspect the pipe run to ensure there are no sags or "bellies" that could collect solids.

D. Correction: All leaks or sags shall be immediately repaired and re-tested until the system is proven to be 100% sound.

3.06 CLEANUP

A. Upon completion, remove all excess materials, tools, and debris from the site. Leave the site in a clean and restored condition.

END OF SECTION

SECTION 22 70 00

SEWERAGE AND DRAINAGE PIPING

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for interior drain, waste, and vent (DWV) piping and the repair of existing exterior sewerage and drainage piping.

1.02 SUBMITTALS

A. Product Data: For each type of pipe, fitting, and repair coupling.

B. For major repairs, submit a brief plan outlining the method of diagnosis, the extent of the proposed repair, and the materials to be used.

1.03 QUALITY ASSURANCE

A. Regulatory Requirements: All work shall comply with local health department regulations and plumbing codes.

B. Installer Qualifications: All plumbing work shall be performed by a licensed plumber.

PART 2 - PRODUCTS

2.01 INTERIOR DRAIN, WASTE, AND VENT (DWV) PIPING

A. Pipe and Fittings: Shall be Schedule 40, solid wall. Material shall be as specified in the SOW or as required to match existing piping. Options include:
   1. Acrylonitrile-Butadiene-Styrene (ABS): Conforming to ASTM D2661.
   2. Polyvinyl Chloride (PVC): Conforming to ASTM D2665.

2.02 REPAIR COMPONENTS

A. Pipe for Repairs:
   1. Exterior Underground: Shall be PVC pipe, SDR 35, conforming to ASTM D3034.
   2. Interior: Shall be Schedule 40 ABS or PVC to match the existing pipe being repaired.

B. Repair Couplings:
   1. Shielded Transition Couplings: For underground repairs or connecting dissimilar pipe materials, use a flexible coupling consisting of a neoprene rubber sleeve and a stainless-steel shear band, conforming to ASTM C1173.
   2. Solvent-Weld Repair Couplings: For interior repairs on ABS or PVC, standard solvent-weld couplings may be used.

2.03 SOLVENT CEMENTS AND PRIMERS

A. General: All solvent cements and primers shall be as recommended by the pipe manufacturer.

B. Primer (for PVC): Shall be a purple-colored primer conforming to ASTM F656.

C. Solvent Cement:
   1. For PVC: Conforming to ASTM D2564.
   2. For ABS: Conforming to ASTM D2235.

PART 3 - EXECUTION

3.01 EXAMINATION AND DIAGNOSIS

A. Investigation: For existing sewer and septic lines reported as having issues, the Contractor shall diagnose the cause and location of leaks, blockages, or other failures.

B. Camera Inspection: Where feasible and necessary, perform a camera inspection of the pipeline to identify the exact location of cracks, root intrusion, collapsed sections, or "bellies" (sags).

C. Report: Submit a brief report or summary of findings to the COR, along with a proposed repair plan, before proceeding with any excavation or major repair work.

3.02 INSTALLATION (INTERIOR DWV)

A. General: Install all new interior drain, waste, and vent (DWV) piping in strict accordance with all applicable plumbing codes.

B. Slope: Install all horizontal drainage piping at a uniform slope to ensure gravity flow. The minimum slope shall be 1/4 inch per foot for pipes 3 inches in diameter or smaller, and 1/8 inch per foot for larger pipes.

C. Joints: Make all solvent-weld joints using the appropriate primer and cement for the pipe material. Ensure joints are clean and fully seated.

D. Support: Securely support all piping with approved hangers or straps to prevent sagging.

3.03 REPAIR PROCEDURE (EXTERIOR/UNDERGROUND)

A. Excavation: Carefully excavate and expose the damaged section of pipe. Shore the excavation as required for worker safety.

B. Removal: Neatly cut out the damaged section of pipe, ensuring the remaining ends of the existing pipe are sound and have clean, square cuts.

C. Installation of New Section:

   1. Install the new section of pipe, ensuring it is aligned with the existing pipe and maintains a continuous, positive slope.
   2. Connect the new section to the existing pipe using the specified shielded transition couplings. Tighten the stainless-steel bands to the manufacturer's recommended torque to create a permanent, watertight seal.

3. Ensure the repaired section is supported by firm, stable bedding material to prevent future settlement.

3.04 BACKFILLING (FOR EXTERIOR REPAIRS)

A. Initial Backfill: After the repair has been tested and approved, carefully place and compact an initial layer of sand or fine gravel backfill to a depth of 6 inches over and around the repaired section to protect it from damage.

B. Final Backfill: Complete the backfilling of the trench, compacting it in lifts to prevent settlement. Restore the surface to its original condition.

3.05 TESTING

A. General: Test all new interior DWV piping and all repaired sections of exterior piping to ensure they are watertight before concealing them (e.g., before closing walls or backfilling trenches).

B. Test Method: Perform a static water test by capping the lower end of the system and filling the pipes with water to a minimum of a 10-foot head. The system shall hold this water for a minimum of 15 minutes with no visible leaks or drop in water level.

C. Correction: All leaks shall be immediately repaired and re-tested until the system is proven to be 100% watertight.

3.06 CLEANUP

A. Upon completion, remove all excess materials, tools, and debris from the site. Leave the site in a clean and restored condition.

END OF SECTION

DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 00 00

HVAC GENERAL REQUIREMENTS

PART 1 - GENERAL

1.01 SUMMARY

A. Includes installation, replacement, and modification of mechanical ventilation and heating equipment, including:
   1. Bathroom exhaust fan/light combination units.
   2. Bathroom ceiling heaters.
   3. Electric wall heaters.
   4. New ductwork and exterior terminations.

B. Incorporates all RFI clarifications including:
   1. Required CFM and sones rating for exhaust fans.
   2. Mandatory exterior venting (no attic discharge permitted).
   3. Acceptable wattage and voltage for electric heaters.
   4. Conditions under which new circuits are required.

1.02 RELATED REQUIREMENTS

A. Division 06 — Blocking for ceiling-mounted devices.
B. Division 09 — Ceiling finishes and drywall repair.
C. Division 22 — Shower valve and bath fixture coordination.
D. Division 26 — Electrical requirements for power supply and new circuits.

1.03 QUALITY ASSURANCE

A. Installers must be licensed HVAC or electrical professionals where required.

B. All installations must comply with current California Building Code and California Mechanical Code.

END OF SECTION

SECTION 23 70 00

MECHANICAL VENTILATION

PART 1 - GENERAL

1.01 SUMMARY

A. Applies to all bathroom fan/light combo replacements across multiple barracks and housing units.

B. Unit quantities shall equal the actual number of bathrooms per building, not the placeholder quantities shown in Attachment 2 (per RFI clarification).

PART 2 - PRODUCTS

2.01 FANS FAN/LIGHT COMBINATION UNITS

A. Units shall meet the following minimum requirements:
   1. Airflow: Minimum 80 CFM (continuous duty).
   2. Sound Rating: Maximum 1.5 sones.
   3. Certification: HVI certified and ENERGY STAR rated.
   4. Lighting: Integrated LED lighting, uniform 3000K warm white color temperature.
   5. Construction: Corrosion-resistant housing and backdraft damper.

2.02 DUCTWORK AND TERMINATIONS

A. Newly installed fan/light units must be vented directly to the building exterior — attic, soffit cavity, or crawlspace terminations are prohibited.

B. Materials:
   1. Smooth-wall metal duct preferred; insulated flex duct permitted where allowed.
   2. Exterior wall caps must include integral backdraft damper and pest screen.

2.03 SWITCHING AND CONTROLS

A. Fans may be switched independently or combined with the light circuit.
B. Humidity-sensing controls are optional unless specified.

PART 3 - EXECUTION

3.01 INSTALLATION OF FAN/LIGHT UNITS

A. Remove existing fan and ductwork.
B. Install new ductwork sized per manufacturer’s requirements.
C. Terminate duct at exterior with proper flashing and sealants.
D. Test unit for sound and airflow performance.

3.02 CEILING REPAIR

A. Patch ceiling surfaces following removal of old units.
B. Finish ceiling per Division 09 requirements.

END OF SECTION

SECTION 23 80 00

HEATING SYSTEMS

PART 1 - GENERAL

1.01 SUMMARY

A. Applies to bathroom ceiling heaters and electric wall heaters identified for replacement in the SOW and RFI clarifications.

PART 2 - PRODUCTS

2.01 BATHROOM CEILING HEATERS

A. Units shall be fan-forced electric ceiling heaters.
B. Wattage: 1,000–1,500 watts, field-verified voltage (120V or 240V).
C. Units must be dedicated heater-only modules unless combination heater/fan/light units are specified.

2.02 ELECTRIC WALL HEATERS

A. Heavy-duty recessed fan-forced heaters for spaces such as the Rock Creek House.
B. Wattage: 1,500–2,000 watts.
C. Powder-coated steel grille.
D. Must match output and performance of existing Cadet heater where replacement is in-kind.

2.03 CONTROLS

A. Wall-mounted thermostat required wall heaters.
B. Ceiling heaters may use wall switch/timer or integral thermostat where permitted.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Remove and dispose of existing heaters.
B. Install new units with all required mounting and electrical connections.
C. Coordinate with Division 26 for circuit requirements.

3.02 ELECTRICAL CAPACITY

A. Replacement heaters may reuse existing circuits only if existing wiring and breaker capacity meet code.

B. If new heater load exceeds available circuit capacity, contractor must notify COR; upgrades handled via contract modification.

END OF SECTION

DIVISION 26 - ELECTRICAL

SECTION 26 00 00

GENERAL ELECTRICAL REQUIREMENTS

PART 1 - GENERAL

1.01 SUMMARY

A. Includes electrical wiring, circuits, grounding, outlets, lighting, and device installations across all buildings included in the Schedule of Work.

B. Incorporates all relevant RFI clarifications, especially for:
   1. Trailer pad RV pedestals (30A / 50A / 20A GFCI).
   2. Electrical capacity verification.
   3. Heater circuit requirements.
   4. New lighting fixture specifications (LED, 3000K).
   5. Exhaust fan/light combination circuits.

1.02 RELATED REQUIREMENTS

A. Division 01 - General Requirements.
B. Division 23 - Mechanical ventilation and heating.
C. Division 22 - Plumbing (coordination for water heater wiring if applicable).
D. Division 31 -Trenching and backfill for conduit runs.

1.03 QUALITY ASSURANCE

A. All electrical work must comply with:
   1. NEC (National Electrical Code).
   2. California Electrical Code.
   3. Local Authority Having Jurisdiction (AHJ) requirements.

B. All wiring must be installed by licensed electricians.

PART 2 - PRODUCTS

2.01 LIGHTING FIXTURES

A. Interior Lighting:
   1. General: All lighting fixtures shall be UL-listed and utilize energy-efficient LED technology.
   2. Selection: Manufacturer, model, and style to be specified in the Statement of Work (SOW).

B. Bathroom Fan/Light Combos:
   1. Shall be a UL-listed, integrated fan and light unit suitable for use in wet locations.

2. Ventilation performance shall be as specified in Section 23 70 00 - Ventilation Equipment.

2.02 WIRING DEVICES

A. Switches and Receptacles (Outlets):
   1. Type: Shall be Specification Grade or better.
   2. GFCI Protection: Provide Ground-Fault Circuit-Interrupter (GFCI) protected receptacles in all locations required by the NEC, including bathrooms, kitchens (countertops), and outdoors.
   3. Color: White, unless otherwise specified in the SOW.

B. RV Power Pedestals (for Trailer Pads):
   1. Type: Shall be a UL-listed, prefabricated RV power pedestal with a rainproof enclosure.
   2. Receptacles: Each pedestal shall contain, at a minimum, one 50-amp (NEMA 14-50R), one 30-amp (NEMA TT-30R), and one 20-amp GFCI (NEMA 5-20R) receptacle, each protected by its own appropriately sized circuit breaker within the pedestal.
   3. Installation: Pedestal shall be securely mounted on a concrete base or as recommended by the manufacturer.

2.03 WIRE, CONDUIT, AND CIRCUIT PROTECTION

A. Interior Wire:
   1. Shall be Type NM-B non-metallic sheathed cable ("Romex"), with conductor sizes appropriate for the circuit amperage (e.g., 14 AWG for 15-amp circuits, 12 AWG for 20-amp circuits).

B. Underground Feeder (UF) Cable:
   1. For Trailer Pads: Shall be Type UF-B (Underground Feeder) direct-burial cable, sized appropriately for the load and distance to the RV pedestal (e.g., 6 AWG copper for a 50-amp service) .

C. Conduit:
   1. Underground: Schedule 40 PVC electrical conduit. Use Schedule 80 PVC where conduit is exposed to potential physical damage.
   2. Exposed (above grade): Rigid Metal Conduit (RMC) or Intermediate Metal Conduit (IMC) where conduit emerges from the ground to protect the cable.

D. Circuit Breakers:
   1. New circuit breakers shall be of the same manufacturer and type as the electrical panel in which they are being installed to ensure proper fit and function.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Verify Conditions: Before beginning work, examine all existing panels, circuits, and substrates. Verify that main electrical panels have sufficient capacity and physical space to accommodate new circuits. Report any conditions that may adversely affect the work to the COR in writing.

3.02 INSTALLATION

A. General: Install all electrical components, wiring, fixtures, and devices in strict accordance with the National Electrical Code (NEC), the manufacturer's written instructions, and all applicable local codes.

B. Interior Work:
   1. Install new lighting fixtures, replacing existing ones as specified in the SOW.
   2. Install new switches, receptacles, and GFCI devices as specified.
   3. Run new branch circuits from the main panel as required for new loads. All wiring in concealed spaces (walls, ceilings) shall be neat and properly supported.

C. Trailer Pad Service Installation:
   1. Trenching and Conduit: Perform all trenching, conduit installation, and backfilling as specified in Section 31 20 00 - Earth Moving.
   2. Cable Installation: Pull the specified UF-B cable through the underground conduit from the main building panel to the pedestal location.
   3. Pedestal Mounting: Securely mount the RV power pedestal on its concrete base at the location specified in the SOW.
   4. Connections: Make all final connections at the pedestal and at the main panel, including the installation of the new, appropriately sized circuit breaker(s).

3.03 TESTING

A. Circuit Testing: After installation and before energizing, test all new and modified circuits for continuity, shorts, and proper grounding.

B. Device Testing: Test every receptacle for correct polarity. Test every GFCI receptacle and GFCI circuit breaker using its integral test button to ensure it trips correctly.

C. Voltage Testing: Test the receptacles at the RV power pedestal to verify correct voltage (e.g., 120V at the 20-amp and 30-amp outlets; 120V/240V at the 50-amp outlet).

D. Functional Test: Turn on all new lighting fixtures and test their corresponding switches to ensure proper operation.

3.04 LABELING AND CLEANUP

A. Panel Labeling: Clearly and legibly label all new circuit breakers in the electrical panel directory (e.g., "Trailer Pad #1," "Bath Lights & Fan").

B. Cleanup: Keep the work area clean daily. Upon completion, remove all scrap wire, packaging, and debris. Clean all fixtures, plates, and devices.

END OF SECTION

SECTION 26 05 00

COMMON WORK RESULTS FOR ELECTRICAL

PART 1 - GENERAL

1.01 SUMMARY

A. Applies to conduit, wiring, grounding, supports, penetrations, and electrical demolition.

PART 2 - PRODUCTS

2.01 CONDUIT AND WIRING

A. Conduit:
   1. Schedule 40/80 PVC for underground runs.
   2. EMT for interior exposed installations.

B. Conductors:
   1. Copper, THHN/THWN or XHHW as required.
   2. Sized per NEC for load.

2.02 GROUNDING

A. Ground rods and bonding jumpers per NEC.

PART 3 - EXECUTION

3.01 TRENCHING AND CONDUIT INSTALLATION

A. Install conduit at depth required by NEC and local code.
B. Place warning tape above buried conduit.

3.02 CIRCUIT VERIFICATION

A. Contractor must verify panel capacity before adding new loads.
B. Existing circuits may be reused only if wire gauge and breaker ratings meet code for new equipment.
C. If panel/heavy load capacity is insufficient, contractor must notify COR.

END OF SECTION

SECTION 26 09 00

CONTROLS

PART 1 - GENERAL

1.01 SUMMARY

A. Applies to thermostat controls for heaters, switches for lighting and fan/light units, and bathroom heater controls.

PART 2 - PRODUCTS

2.01 CONDUIT AND WIRING

A. Thermostats: Line-voltage wall thermostats for electric wall heaters.
B. Switches: Standard toggle, rocker, or timer switches.
C. All controls must comply with NEC and manufacturer requirements.

PART 3 - EXECUTION

A. Install switches at standard mounting heights unless existing conditions dictate otherwise.
B. Coordinate circuits with mechanical equipment.

END OF SECTION

SECTION 26 50 00

LIGHTING

PART 1 - GENERAL

1.01 SUMMARY

A. Applies to replacement of interior lighting fixtures, vanity lights, and integrated LED fixtures in multiple buildings.

PART 2 - PRODUCTS

2.01 INTERIOR LIGHTING FIXTURES

A. All fixtures must be:
   1. Integrated LED (no screw-in LED bulbs allowed).
   2. 3000K color temperature.
   3. UL-listed for residential or light commercial use.

B. Fixture Types:
   1. Surface-mount or semi-flush ceiling fixtures.
   2. Recessed lay-in LED panels (for grid ceilings).
   3. Wall-mounted LED vanity bars (minimum 24 inches, 3-lamp equivalent).

2.02 BATHROOM FAN/LIGHT COMBO FIXTURES

A. Must comply with Division 23 airflow and sound requirements.

PART 3 - EXECUTION

A. Remove old fixtures and patch surfaces as needed.
B. Install new fixtures with proper mounting hardware.
C. Test lighting circuits and ensure uniform light output.

END OF SECTION

SECTION 26 60 00

ELECTRICAL SERVICE FOR TRAILER PADS

PART 1 - GENERAL

1.01 SUMMARY

A. Applies to electrical service pedestals for new trailer pads at Mammoth Ranger District Compound, Crestview Fire Station Compound and Lee Vining Ranger Station Compound.

PART 2 - PRODUCTS

2.01 RV POWER PEDESTALS

A. Each pedestal must include:
   1. 50A NEMA 14-50R outlet.
   2. 30A NEMA TT-30R outlet.
   3. 20A GFCI duplex outlet.
   4. Individual internal circuit breakers.

2.02 CONDUIT AND CONDUCTORS

A. Underground conduit: Schedule 40 PVC minimum.
B. Copper conductors sized per NEC for 50A/30A circuits.

2.03 GROUNDING

A. Install grounding electrodes systems per NEC.

PART 3 - EXECUTION

3.01 PANEL CONNECTIONS

A. Contractor’s scope begins at breaker termination inside the main panel.

B. Contractor must verify:
   1. Available breaker spaces.
   2. Panel load capacity.
   3. Existing wire sizing and routing.

3.02 TRENCHING AND INSTALLATION

A. Install conduit in trenches with proper depth and bedding.
B. Pull conductors and terminate at pedestal and panel.

3.03 TESTING AND COMMISIONING

A. Test pedestal voltage and grounding integrity.

B. Ensure GFCI devices function properly.

END OF SECTION

DIVISION 31 - EARTHWORK

SECTION 31 00 00

EARTHWORK

PART 1 - GENERAL

1.01 SUMMARY

A. Includes all excavation, trenching, backfilling, grading, and compaction as required for:
   1. Trailer pad preparation (concrete pads, utility connections).
   2. Water and sewer line installation.
   3. Electrical conduit installation.
   4. Repairs to disturbed soils and replacement of unsuitable material.

B. Incorporates all relevant RFI clarifications including:
   1. Minimum excavation depth for utilities.
   2. Subgrade preparation for concrete pads.
   3. Import/export soil requirements.
   4. Compaction standards (95%/90% depending on area).
   5. Acceptable handing of unknow site conditions.

1.02 RELATED REQUIREMENTS

A. Division 03 - Concrete (Subgrade conditions for pads).
B. Division 22 - Plumbing (Utility line installation).
C. Division 26 - Electrical (Conduit trenching requirements).

1.03 FIELD CONDITIONS

A. Contractor must verify existing utility locations prior to excavation (use 811 or equivalent).
B. Unknown buried conditions discovered during excavation must be immediately reported to COR.
C. Depths of existing sewer laterals at trailer pads sites are unknown; potholing is required (per clarification).

END OF SECTION

SECTION 31 20 00

EARTH MOVING

PART 1 - GENERAL

1.01 SUMMARY

A. This section specifies requirements for excavation, grading, trenching, backfilling, and compaction for the preparation of trailer pads and the installation of underground utilities.

1.02 QUALITY ASSURANCE

A. Regulatory Requirements: Comply with all local regulations and OSHA requirements for excavation safety, including shoring and trench protection.

PART 2 - PRODUCTS

2.01 SOIL MATERIALS

A. Sub-base Material (for Trailer Pads):
   1. Shall be an aggregate base course (ABC) or crushed stone, well-graded, with a maximum particle size of 1-1/2 inches and containing no organic material, clay, or other deleterious substances.

B. Bedding Material (for Utilities):
   1. Shall be clean, well-graded sand or fine gravel with a maximum particle size of 3/4 inch.

C. Backfill Material:
   1. Initial Backfill (over utilities): Shall be the same as the Bedding Material.
   2. Final Backfill: May be the native excavated material, provided it is free of large rocks (over 4 inches in diameter), frozen clumps, organic matter, and construction debris. If native material is unsuitable, import clean fill material as approved by the COR.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Utility Location: Before beginning any excavation, the Contractor shall contact the local utility locating service ("Call Before You Dig") and have all existing underground utilities located and marked. The Contractor is responsible for protecting all existing utilities from damage.

B. Site Conditions: Verify existing grades and site conditions. Report any discrepancies or conditions that may affect the work to the COR before proceeding.

3.02 EXCAVATION AND GRADING (FOR TRAILER PADS)

A. Excavation: Excavate areas for trailer pads to the lines, grades, and depths required to accommodate the specified thickness of the sub-base material and the concrete slab.
   1. Trailer pad excavation depth must allow for:
      a) 4-inch compacted aggregate base course (ABC) sub-base.
      b) 6-inch reinforced concrete slab (per Division 03).

B. Subgrade Preparation: After excavation, compact the native subgrade (the bottom of the excavated area) to a minimum of 90% of its maximum dry density. The subgrade must be firm, stable, and free of any organic material or debris.

C. Over excavation must be filled with engineered fill and compacted.

3.03 SUB-BASE INSTALLATION

A. Placement: Place the specified sub-base material over the prepared subgrade.

B. Compaction:
   1. For trailer pads install the sub-base material in lifts not exceeding 6 inches in loose thickness. Compact each lift to a minimum of 95% of its Standard Proctor Maximum Dry Density (ASTM D698).
   2. For walkways and ramps, compact to 90% maximum dry density.

C. Final Grade: The final compacted sub-base surface shall be smooth, uniform, and graded to the elevations required for the concrete slab, ready to receive formwork.

D. Remove organic soil and unsuitable materials.

3.04 TRENCHING (FOR UTILITIES)

A. Excavation: Excavate trenches for all underground utilities (water, sewer, electrical) to the depths and slopes specified in their respective divisions (Divisions 22 and 26). Trenches shall be wide enough to allow for proper installation and compaction of backfill.

B. Water Lines:
   1. Maintain positive slope per RFI callback.
   2. Provide minimum 4-inch bedding layer.

C. Sewer Lines:
   1. Minimum 2% slope.
   2. Depth varies; contractor must pothole to confirm.
   3. Provide cleanouts per plans.

D. Electrical Conduit:
   1. Install per NEC depth requirements.
   2. Provide warning tape above conduit.

E. Safety: Comply with all OSHA requirements for trench safety, including shoring or sloping of trench walls for any excavation 5 feet or deeper.

F. Trench Bottom: The bottom of the trench shall be stable and free of sharp rocks or debris.

3.05 UTILITY BEDDING AND BACKFILLING

A. Bedding: Place a minimum of 4 inches of the specified bedding material in the bottom of the trench and grade it to provide continuous support for the utility pipe or conduit.

B. Initial Backfill: After the utility has been installed and tested, carefully place and compact initial backfill (bedding material) around and over the utility to a depth of 6 inches above the top of the pipe/conduit.

C. Warning Tape: Place the specified warning tape in the trench, approximately 12 inches above the top of the utility line.

D. Final Backfill: Place and compact the final backfill material in lifts not exceeding 8 inches in loose thickness. Compact each lift to a minimum of 90% of its maximum dry density.

3.06 DISPOSAL OF EXCESS MATERIAL

A. The Contractor is responsible for legally transporting and disposing of all excess excavated soil and debris off-site.

3.07 FINAL GRADING AND SURFACE RESTORATION

A. Upon completion of all backfilling, grade all disturbed areas to blend smoothly with the surrounding landscape.

B. Ensure final grading provides positive drainage away from all trailer pads and buildings.

C. Restore all surface areas, including any lawns or paved areas, to their original condition or better.

END OF SECTION

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 00 00

EXTERIOR IMPROVEMENTS

PART 1 - GENERAL

1.01 SUMMARY

A. Includes all exterior work not covered under structural, utility, or architectural divisions, including:
   1. Final grading around new trailer pads.
   2. Gravel surfacing.
   3. Site cleanup and restoration.
   4. Minor site repairs associated with utility installation.

B. Incorporates all applicable RFI clarifications regarding:
   1. Gravel placement and depth.
   2. Restoration of disturbed areas.
   3. Transitions from new concrete pads to adjacent grade.
   4. Sloping requirements for drainage.

1.02 RELATED REQUIREMENTS

A. Division 31 - Earthwork.
B. Division 22 - Water and Sewer Installation.
C. Division 26 - Electrical Conduit and Pedestal Installations.

PART 2 - PRODUCTS

2.01 AGGREGATE MATERIALS

A. Gravel surfacing to be:
   1. 3/4-inch crushed rock or decomposed granite (DG), local source acceptable.
   2. Free of clay, organics, and debris.

2.02 GEOTEXTILES

A. Non-woven geotextile fabric for separation under gravel where soils are loose or soft.

PART 3 - EXECUTION

3.01 GRADING

A. Grade areas adjacent to concrete trailer pads provide positive drainage away from structure at minimum 2% slope.

B. Ensure smooth transition between pad edges and natural grade.

3.02 GRAVEL PLACEMENT

A. Install 3 inches minimum of compacted gravel or DG in areas indicated in the SOW.

B. Gravel may be used around trailer pads where soil is disturbed by trenching operations.

C. RFI clarification: Gravel placement is not required over entire sites—only where shown or affected by construction.

3.03 RESTORATION OF DISTURBED AREAS

A. Restore all disturbed surfaces to pre-construction condition or better.
B. Remove construction debris, abandoned materials, surplus soil, and spoils.
C. Re-grade ruts or depressions caused by equipment.

3.04 SITE CLEANUP

A. Perform final cleanup of project area including removal of trash, debris, vegetation cuttings, and equipment.

B. Leave site in clean, safe, and operable condition.

END OF SECTION

ADDENDUM

1. GENERAL PROJECT-WIDE CLARIFICATIONS

A. Verification of Existing Conditions
   1) Contractors are required to field-verify all dimensions, quantities, and existing conditions prior to ordering materials.
   2) Discovery of concealed conditions outside the anticipated scope must be reported to the COR for direction prior to proceeding.

B. Occupied Facilities Clarification
   1) Several buildings will remain fully or partially occupied during construction.
   2) Contractor must provide:
      a) Minimum 14 calendar days' notice before disruptive work begins.
      b) Temporary egress always.
      c) Dust and noise control per Division 01.

C. Lead and Asbestos Assumptions
   1) No project-wide hazardous materials assessment exists.
   2) Buildings constructed prior to 1978 must be treated as possibly containing lead-based paint (LBP) or asbestos-containing materials (ACM).
   3) Contractor must follow EPA RRP practices and stop work immediately if suspect ACM is encountered.

2. TRAILER PAD & UTILITIES CLARIFICATIONS (ALL SITES)

A. Trailer Pad Dimensions and Construction
   1) All trailer pads must be 12 ft x 25 ft x 6 in thick concrete slabs with #4 rebar at 12" O.C. each way.
   2) Subgrade must be compacted to 95% maximum dry density.

B. RV Pedestal Requirements
   1) Each pedestal must include:
      a) 50A (14-50R) receptacle
      b) 30A (TT-30R) receptacle
      c) 20A GFCI duplex outlet
      d) Internal breakers

C. Utility Connection Boundaries
   1) Contractor scope begins at existing main distribution lines (electrical, water, sewer).
   2) Unknown sewer line depths require potholing.

3. INTERIOR UNIT COUNTS & FIXTURE REQUIREMENTS

A. Bathroom Fan/Light Units
   1) Provide one replacement for each bathroom. Must meet:
      a) ≥80 CFM
      b) ≤1.5 sones
      c) ENERGY STAR rating

4. Window Quantity Confirmation (Crestview Cabins)

A. Cabin 1119: 10 windows
B. Cabin 1120: 8 windows
C. Cabin 1122: 8 windows

5. Flooring Clarifications

A. Carpet tile only; broadloom carpet prohibited.
B. LVP only in dry areas; glue-down installation.
C. Sheet vinyl in bathrooms must have heat-welded seams.

6. REQUIRED GOVERNMENT REVIEWS / SUBMITTALS

A. Minimum 10 working days are required for Government review of submittals.
B. Contractor must include product data, color charts, and warranty documentation.

7. GENERAL ROOFING CLARIFICATIONS (ALL SITES)

A. Full Roof Tear-Off Requirement
   1) All reroofing areas require complete removal down to bare decking, including:
      a) Shingles or membrane material
      b) Underlayment.
      c) Flashings and vents
      d) Drip edges
      e) Damaged decking (per COR direction)
      f) Existing gutters remain unless noted otherwise; downspout replacements are only as specified.

8. ASPHALT SHINGLE ROOFING CLARIFICATIONS

A. Architectural Shingle Requirement (RFI Confirmed)
   1) Architectural laminated shingles are mandatory on all steep-slope roofs.
   2) 3-tab shingles are not acceptable under any circumstance.

A. Shingle Performance Requirements Updated
   1) Shingles must meet:
      a) ASTM D3462
      b) Class A fire rating
      c) Wind rating 110 mph minimum, upgradeable to 130 mph with enhanced fastening
      d) Algae resistance: 10–15-year warranty minimum

9. UNDERLAYMENT & ICE-AND-WATER SHIELD (IWS) CLARIFICATIONS

A. Standard Roof Slopes (>4:12). IWS required at:
   1) All eaves, extending 36 in. inside heated wall line
   2) Valleys and penetrations
   3) Synthetic underlayment required over remaining deck.

B. Low-Slope Roofs (2:12 to 4:12) — Critical RFI Change
   1) Entire roof surface must be covered with FULL Ice-and-Water Shield membrane.
   2) Synthetic or felt-only underlayment systems are prohibited.

C. Very Low Slopes (≤1/2:12)
   1) These areas (e.g., Helitack Office Shed connector roof) must receive SBS-modified bitumen membrane systems, 2- or 3-ply.

10. DECKING REPLACEMENT & STRUCTURAL ITEMS

A. Included Deck Replacement Allowance
   1) Contractor must include an allowance of three (3) sheets of plywood per building.
   2) Additional decking requires COR verification and modification.

B. Structural Repairs
   1) If bowed, rotted, or split rafters/studs are encountered, contractor must notify COR.
   2) Some locations (e.g., Crestview Cabin 1120) require engineered structural repair plans.

11. ROOF FLASHINGS & METAL COMPONENT CLARIFICATIONS

A. Drip Edge
   1) All drip edge material must be:
      a) 24-gauge steel or heavy-duty aluminum.
      b) Dark Brown prefinished enamel (mandatory across all sites).

B. Flashing Requirements
   1) Replace all step, counter, and valley flashings.
   2) Ridge vents must be continuous high-flow type.

C. Pipe Boots
   1) All new roof penetrations require new neoprene or lead boots.

12. METAL ROOFING CLARIFICATIONS (Crestview & Lee Vining)

A. Metal Roof Replacement Items
   1) Provide standing seam or equivalent system where indicated.
   2) Match existing panel profile, color, and finish as closely as possible.
   3) Snow Retention System (Required).
      a) All metal roofs must include a continuous snow retention system, engineered for site-specific loads.
      b) Fasten into structural framing members.

13. HISTORIC SHAKE ROOF (1025 Lee Vining Small House)

A. Material Requirements
   1) Western Red Cedar shakes: Certi-Split Premium Grade, hand-split, resawn, edge-grain.
   2) Fire-retardant treated (Certi-Guard).
   3) Assembly must meet Class B minimum rating (Class A with approved underlayment system).

B. Installation Clarifications
   1) Must match existing spacing, coursing, and exposure.
   2) Coordinate work with SHPO-related requirements.

14. WARRANTY CLARIFICATIONS

A. Asphalt Shingles
   1) Provide Lifetime Limited Warranty with 20-year non-prorated protection.
   2) Include wind uplift and algae protection.

A. Low-Slope Membranes
   1) Provide 10-year material warranty (NDL not required).
   2) Provide 2-year contractor workmanship warranty.

15. GENERAL OPENINGS CLARIFICATIONS (ALL SITES)

A. Verification of Existing Openings
   1) Contractor must field-measure all window and door openings prior to ordering materials.
   2) No additional compensation will be made for incorrect field sizes due to lack of verification.
   3) Window and door counts shown in Attachment 2 were placeholders; updated verified counts apply.

16. WINDOW CLARIFICATIONS

A. Frame Material Requirement
   1) All replacement windows must be VINYL FRAME ONLY.
   2) Aluminum windows are prohibited due to condensation, thermal transfer, and energy performance.

B. Window Operation Type
   1) Required style: Horizontal sliding windows (single-slider configuration).
   2) One sash operable, one sash fixed.

C. Energy Performance Requirements
   1) Glazing must meet:
      a) Low-E insulated glazing.
      b) Argon gas fill.
      c) U-factor ≤ 0.30.
      d) SHGC ≤ 0.25.

D. Tempered Glazing Requirements
   1) Tempered glazing is required:
      a) In bathrooms.
      b) In windows adjacent to doors (within code distance).
      c) At low sill heights (meeting California Building Code hazard criteria).

E. Egress Requirements (Bedrooms)
   1) Windows in bedrooms must meet current egress size and operability criteria.
   2) If existing framed opening is too small to accommodate an egress window, contractor must notify COR.

F. Updated Window Counts (Crestview Cabins)
   1) These counts supersede those shown in Attachment 2:
      a) Cabin 1119 - 10 windows.
      b) Cabin 1120 - 8 windows.
      c) Cabin 1122 - 8 windows.

G. Accessories & Trim Requirements
   1) All operable windows must include insect screens.
   2) Exterior flashing tape and WRB integration must meet manufacturer requirements.
   3) Interior trim must be reinstalled or replaced for a complete finish.

17. DOOR CLARIFICATIONS (EXTERIOR & INTERIOR)

A. Exterior Doors Must Be Prehung Units.
1) All exterior doors must be complete prehung assemblies, including:
   a) New frame.
   b) Hinges.
   c) Threshold.
   d) Weatherstripping.
   e) Hardware prep.
   f) Slab-only replacements are not allowed.

B. Exterior Door Material Requirements.
   1) Housing/barracks units: Insulated fiberglass doors.
   2) Service rooms/utility areas: Insulated steel doors (24-gauge min.).

C. Interior Doors
   1) Bathrooms: Solid core.
   2) Bedrooms/closets: Hollow core permitted if existing frames are sound.
   3) Replace casing/trim as required for complete installation.

D. Weatherstripping Clarifications
   1) All exterior doors must receive new adjustable compression weatherstripping.
   2) Two (2) primary exterior doors in Mammoth Fire Barracks specifically require weatherstripping upgrades.

E. Door Hardware Requirements
   1) All exterior locksets must be SFIC-compatible (Small Format Interchangeable Core).
   2) Contractor provides temporary construction cores; Government provides final cores.
   3) Hardware finish must be consistent (Satin Nickel or approved equal).

18. STORM & SCREEN DOOR CLARIFICATIONS

A. Storm Door Requirements
   1) Heavy-duty aluminum frame.
   2) Interchangeable glass and screen inserts.
   3) Pneumatic closer and matching hardware.

B. Screen Door Requirements
   1) Rigid aluminum or wood frame.
   2) Heavy-duty fiberglass or aluminum mesh.
   3) Self-closing hinges.

19. GENERAL INTERIOR CLARIFICATIONS

A. Occupied Facilities - Interior Work Protocols

1) Many buildings remain occupied; contractor must:
2) Provide 14-day advance notice before disruptive work.
3) Install dust partitions and floor protection.
4) Always maintain safe access and emergency egress.

B. Lead-Safe Work Practices
1) All pre-1978 structures must be treated as potentially containing lead paint.
2) HEPA vacuuming, containment, and surface cleaning required.

20. DRYWALL, CEILING, AND WALL FINISH CLARIFICATIONS

A. Gypsum Board Replacement Requirements
1) Bathrooms: Replace with 5/8" moisture- and mold-resistant gypsum board.
2) Ceilings: Install 5/8" Type X at all ceiling tile removal locations.

B. Ceiling Tile to Drywall Conversion
1) Where ceiling tiles are removed, contractor must:
a) Remove entire suspension grid.
b) Install new framing or furring for a continuous substrate.
c) Finish ceilings to a smooth Level 4 drywall finish (no texture allowed).

C. Wall Texture Clarification
1) Walls may receive light orange-peel texture only where matching existing conditions.
2) Ceilings must remain smooth.

21. FLOORING CLARIFICATIONS

A. Carpet Tile
1) Carpet tile ONLY allowed; broadloom carpet is prohibited.
2) Required characteristics:
a) Minimum 20 oz/yd² face weight.
b) Solution-dyed nylon fiber.
c) Fiberglass-reinforced backing (e.g., Shaw EcoWorx).
d) Installation: Direct glue-down, using pressure-sensitive adhesive.
e) No carpet pad allowed.

B. Luxury Vinyl Plank (LVP) Flooring
1) LVP only allowed in dry areas (kitchens, living rooms, hallways).
2) Minimum requirements:
a) 3.0 mm thickness.
b) 20 mil wear layer.
c) Glue-down installation only (floating floors prohibited).
d) Provide transitions and new vinyl cove base.

C. Sheet Vinyl in Bathrooms
1) Minimum 2.0 mm thickness and 20 mil wear layer.
2) Heat-welded seams required for waterproofing.
3) Subfloors must be leveled and prepared to eliminate telegraphing.
4) Substrate Preparation Requirements
5) Remove old adhesives fully.
6) Grind or level uneven floor surfaces.

7) Secure loose subfloor panels and eliminate squeaks.

22. PAINTING & COATING CLARIFICATIONS

A. Interior Paint System Requirements
1) Ceilings: Flat or ultra-flat finish.
2) Walls (dry areas): Eggshell or satin.
3) Walls (bathrooms/kitchens): Semi-gloss.
4) Trim/doors/baseboards: Semi-gloss.

B. Exterior Paint System Requirements
1) Three-coat system required:
2) Primer
3) Two finish coats
4) Sheens:
5) Siding/soffits: Satin or low-luster.
6) Trim/doors: Semi-gloss.

C. Surface Prep Clarifications
1) All minor cracks, holes, and blemishes must be patched.
2) Large surfaces requiring repair fall under drywall/siding scope.

A. Matching Existing Colors
1) Colors to match existing unless otherwise directed by the Government.
2) No accent walls required.

23. CEILING TILE REPLACEMENT CLARIFICATIONS

A. New Tile Requirements
1) Ceiling tile replacement is in-kind, unless ceilings are being converted to drywall.
2) Minimum standards:
a) Class A fire rating (ASTM E84).
b) NRC $\geq$ 0.55.
c) 2×2 or 2×4 tiles; entire room replaced for consistency.

B. Grid Repair
1) Damaged suspension grids must be repaired and leveled before new tiles are installed.

24. GENERAL PLUMBING CLARIFICATIONS (ALL SITES)

A. Scope Boundary Clarification
1) Contractor's plumbing work begins at the existing supply and waste lines within each building unless noted otherwise.
2) For trailer pads, the contractor's scope begins at the existing main water and sewer distribution points, confirmed by potholing when depths are unknown.

B. WaterSense Requirements
1) All plumbing fixtures must meet EPA WaterSense performance standards, including:
a) Maximum 1.28 GPF toilets.
b) 1.5 GPM lavatory faucets.
c) 1.8 GPM shower heads.

C. Fixture Replacement Protocols
   1) All plumbing fixtures identified for replacement must be removed and replaced in full, including all supply lines, escutcheons, stops, and trim components.

25. SHOWER SYSTEM CLARIFICATIONS

A. Shower Valve Requirement
   1) Replacement of the entire valve body is required, not just trim.
   2) New valves must be pressure-balancing or thermostatic mixing types meeting current code.

B. Shower Head & Trim
   1) Maximum flow rate 1.8 GPM.
   2) New arm, flange, and trim kit required.

C. Shower Surround Removal & Wall Repair
   1) Existing surrounds must be removed down to framing if replacement is indicated.
   2) Replace damaged studs or backing as required.
   3) Install 5/8" moisture-resistant gypsum board before new surround.

26. SINKS, FAUCETS & VANITIES CLARIFICATIONS

A. Bathroom Sinks & Faucets
   1) Sinks must be vitreous china, under-mount or drop-in depending on vanity configuration.
   2) Lav faucets must meet 1.5 GPM max flow.
   3) Install new supply stops, stainless braided lines, and P-traps.

B. Kitchen Sinks & Faucets
   1) Minimum 18-gauge stainless steel double-bowl sink.
   2) Kitchen faucets must be 1.8 GPM, single-handle, with new supply lines and stops.

C. Vanities (Coordination with Div 06)
   1) Replace in kind (wall-hung for wall-hung, floor-mounted for floor-mounted).
   2) Provide new blocking for wall-hung units.
   3) Replace countertops with solid surface or cultured marble with integrated bowl.

27. TOILET REPLACEMENT CLARIFICATIONS

A. Toilet Specifications
   1) Two-piece vitreous china construction.

28. GENERAL ELECTRICAL CLARIFICATIONS (ALL SITES)

A. Panel Capacity & Circuit Verification
   1) Contractor must verify available breaker capacity, wire gauge, and panel load before adding any new circuits.
   2) Existing circuits may be reused only if they meet current NEC requirements for:
      a) Conductor sizing.
      b) Breaker rating.
      c) Load capacity / derating.

3) If panel capacity is insufficient, contractor must notify COR; electrical upgrades require contract modification.

B. Trenching Requirements for Electrical Conduit
1) Conduit for trailer pad pedestals must be buried at NEC-required depth (minimum 18–24 inches depending on local code).
2) Warning tape must be installed above buried conduit.
3) Bedding must consist of clean sand or native material free of sharp stone.

29. RV POWER PEDESTALS CLARIFICATIONS

A. Pedestal Configuration
1) Each trailer pad must receive a fully wired RV pedestal including:
a) 50A NEMA 14-50R receptacle.
b) 30A NEMA TT-30R receptacle.
c) 20A GFCI duplex receptacle.
d) Individual internal circuit breakers.

B. Circuit Requirements
1) Conductors must be copper and sized per NEC for 50A & 30A circuits.
2) Grounding electrode system required at each pedestal.
3) Contractor scope begins at breaker termination inside the main panel.

C. Voltage & Function Testing
1) All pedestals must be tested for:
a) Voltage accuracy (120V/240V).
b) Proper grounding.
c) GFCI function.

30. EXHAUST FAN/LIGHT COMBINATION CIRCUIT CLARIFICATIONS

A. Unit Replacement Requirement
1) Provide one new fan/light combo per bathroom, per corrected building counts.

B. Electrical requirements:
1) 80 CFM minimum.
2) ≤1.5 sones.
3) Integrated LED lighting.
4) Fans may share a circuit with bathroom lighting only if circuit load permits.

C. New Ducting & Exterior Termination
1) Fans may not terminate in attic, soffit cavity, or crawlspace.
2) Must be ducted to exterior wall with proper flashing and backdraft damper.

31. LIGHTING FIXTURE CLARIFICATIONS

A. Integrated LED Requirement
1) All replacement interior fixtures must be integrated LED (no screw-in retrofits).
2) Color temperature must be 3000K warm white.
3) Fixtures must be UL-listed and suitable for residential/barracks use.

B. Vanity Lighting Requirements
   1) Minimum 24-inch LED bar fixture for bathrooms.

C. Replace damaged junction boxes and repair drywall as required.

32. HEATING EQUIPMENT ELECTRICAL CLARIFICATIONS

A. Electric Wall Heaters
   1) New units must match wattage & capacity of existing Cadet heaters unless otherwise directed.
   2) Typical range: 1,500–2,000 watts.

A. Bathroom Ceiling Heaters

33. GENERAL SITEWORK & UTILITY CLARIFICATIONS (ALL SITES)

A. Scope Boundaries Clarified
   1) Contractor responsibility for site utilities begins at existing main distribution points (electrical, water, and sewer).
   2) Contractor must pothole to confirm depth and alignment of existing utilities before trenching or installation of trailer pad connections.
   3) Unknown utility depths cannot be assumed; discovery must be verified with COR.

A. Excavation & Backfill Requirements
   1) All trenching must follow NEC, UPC, and project specifications for depth, slope, and bedding.
   2) Backfill may not occur until utilities are pressure-tested, leak-tested, or electrically inspected.

34. WATER SYSTEM CLARIFICATIONS

A. Water Distribution Piping
   1) Accepted piping materials (per RFI):
      a) Type L copper, or
      b) PEX-A (with appropriate crimp/expansion fittings)

B. All water lines must maintain positive slope to prevent trapped water and freeze hazards.

C. Bedding & Cover Requirements
   1) Provide 4 inches minimum clean sand bedding beneath water lines.
   2) Provide 6 inches minimum cover above pipe before general backfill.

D. Freeze Protection Requirements
   1) Maintain proper depth per local conditions.
   2) Insulation or heat tape required only where existing conditions necessitate.

35. SEWER SYSTEM CLARIFICATIONS

A. Minimum Slope
   1) All sewer piping must maintain 2% minimum slope (1/4 inch per foot).

B. Potholing Requirement

1) Existing sewer laterals at trailer pad locations have unknown depths.
2) Contractor must pothole to identify actual depth and confirm connection feasibility.

C. Bedding Requirements
1) Provide 4 inches minimum sand bedding beneath sewer piping.
2) Provide uniform support along entire pipe length.

D. Cleanouts
1) Install cleanouts per drawings and code.
2) Ensure cleanouts terminate at grade unless otherwise noted.

36. ELECTRICAL TRENCHING & SITE UTILITIES (Cross-Reference)

A. Trench Depth Requirements
1) Follow NEC burial depths (18–24 inches depending on conduit type and location).
2) Provide warning tape 12 inches above conduit, continuous along trench length.

B. Conduit Type
1) Schedule 40 PVC required underground; Schedule 80 where exposed.
2) Backfill Requirements
3) Trench backfill must occur in compacted lifts (maximum 8 inches).

C. Compaction minimum:
1) 90% for utility trenches.
2) 95% for trailer pads and structural areas.

37. TRAILER PAD CONSTRUCTION CLARIFICATIONS

A. Concrete Pad Requirements
1) Trailer pad dimensions: 12 ft × 25 ft × 6 in thick concrete slab.
2) Reinforcement: #4 rebar at 12 in on center, each way.
3) Subgrade must be compacted to 95% maximum dry density.

B. Gravel Around Trailer Pads
1) Gravel installation is required only where:
a) Trenching disturbed soil, or
b) Shown in the Schedule of Work.

C. Minimum gravel depth: 3 inches compacted.

D. Site Restoration Requirements
1) Restore all disturbed areas to pre-construction condition or better.
2) Remove spoils, surplus soils, debris, and construction materials.
3) Smooth transitions between pad edges and natural grade are required.

38. EARTHWORK CLARIFICATIONS

A. Excavation Requirements
1) Excavate only to required depth for utilities or pad sections.
2) Over-excavation must be filled with engineered fill and compacted.

B. Soil Suitability
   1) Remove organic or unsuitable soils.
   2) Haul-off unsuitable materials must follow COR direction.

C. Compaction Standards
   1) Trailer pad areas: 95% max dry density.
   2) Walkways/ramps: 90% max dry density.
   3) Utility trench backfill: 90% unless noted otherwise.